



The Emergence of Cyber Threat as a Global Risk

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Abstract

This paper studies the role of cyber risk in the disruption of the world peace. It goes through the analysis of the physical and cyber wars relationship, their economic impacts on the involved countries as well as their dyadic determinants. Placing ourselves in the current geopolitical context, this paper also tries to make some forecasts of the next dyadic conflicts. To do so, multivariate along with parametric and non-parametric models are fitted, such as VAR models, rare-events logit and ensemble methods. Results show self-reinforcing dynamics between cyber and physical conflicts at multiple levels of analysis. On the economic side, cyberattacks are found to be positively associated with GDP per capita once physical conflicts are accounted for. Finally, conflict persistence is found to be significantly more predictable than onset, with cyber variables emerging as key predictors of ongoing conflicts.

Keywords: Conflicts, Cyber-risks, Economics, Forecasts

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1 Introduction

The World War III is about to begin. Indeed, in recent years, an escalation of conflicts has been observed. A non-exhaustive list of examples of such violent events are: the Ukrainian-Russian conflict initiated on February 24th 2022, the Afghanistan-Pakistan one on October 9th 2025 and more recently, the Iran-USA war that started on February 28th 2026. Nowadays, geopolitical instability is one of the major concern. However we observe an evolution in the nature of conflicts, from physical to digital, triggered by the hidden role of cyber threat.

Geopolitical instability and cybersecurity are more than ever at the heart of major global risks. Indeed, according to the future risks reports from [AXA \(2025\)](#), they are part of the Top 3 Risk in 2025 (climate risk is Top 1). The economic impact of such macro-level conflict can be enormous for countries. For instance, according to Time Magazine, “the United States has spent at least \$12 billion on its war with Iran in just the first two weeks of the conflict”.

This anxious current climate gives food for thought, and a central question emerges: how do cyberattacks reshape the dynamics, consequences, and predictability of modern conflicts? These three dimensions are deeply connected: cyber and physical conflicts mutually reinforce each other, this escalation generates measurable economic costs, and these combined dynamics influence which country pairs are most at risk of future confrontation.

Three questions arise naturally from this logic: what are the bidirectional dynamics between physical and cyber conflicts, what are their joint economic consequences, and can conflict be predicted at the dyadic level?

This report is divided into 8 main parts. First a literature review of the three axes of study, namely geopolitics, economics and cyber threat. This is followed by a presentation of the data and an overview of the empirical strategy. First, we study the dynamic interactions between cyberattacks and physical conflicts. Second, we examine their economic consequences. Third, we investigate whether cyber variables improve the prediction of conflicts at the dyadic level. Finally, results are presented before concluding.

2 Literature

To date, there is a clear gap in the literature that simultaneously integrates the three research domains: economics, cybersecurity, and geopolitical conflicts. Addressing this gap will constitute the main, highly significant contribution of this work. While studies exist that examine pairs of these areas, namely economics and geopolitics, geopolitics and cyberconflicts, or cyberconflicts and economics. Few, if any, provide a comprehensive analysis that combines all three dimensions. By bridging this gap, this research aims to offer new insights into the complex interplay between economic factors, cyber threats, and geopolitical tensions.

2.1 Conflict literature

Armed conflicts have been extensively studied in geopolitical science and international relations, with a focus on understanding what drives their onset, duration, and recurrence.

Dyad characteristics and probability of conflicts [Bremer \(1992\)](#) studied this in his paper “Dangerous Dyads: Conditions Affecting the Likelihood of Interstate War”. The author finds the main country’s characteristics increasing the probability of interstate wars to be the contiguity of the countries (i.e. if the countries share a common land border or if they are separated by a short distance), along with the power parity and major power. This findings were confirmed and extended later on by [Oneal and Russett \(1999\)](#) in his work “The Kantian Peace: The Pacific Benefits of Democracy, Interdependence, and International Organizations”. The authors base their study on the Kantian theory, according to which a democratic country involved in international organizations and international trades has a high probability of peace and therefore not being involved in conflicts. Indeed, trade, democracy and membership in international organizations significantly decrease the probability of conflict, along with complementary variables such as the balance of power.

Temporal dimension of conflicts [Goertz and Diehl \(1993\)](#) show a different kind of conflict study. Indeed, their goal was not to work on what could lead to wars but whether on identifying if conflicts are related across time. They highlight the historical perspective between interstate conflicts and show they are not simply isolated events. More precisely, the heart of the work done

in this paper is to define enduring rivalries. They are in opposition to the one derived from rational choice literature, which is based on game theory models. Here, they define enduring rivalries as the combination of competitions with a temporal and spatial perspectives. Indeed, enduring rivalries are over different connected competitions, for a long period of time (decades), between recurrent states (Goertz and Diehl, 1993). They introduce the idea that wars can be linked one to another and that the rivalry between several states can last for years, leading to several conflicts. This describes long run relationship.

2.2 Cybersecurity literature

Cyberattacks play a central role in the analysis of international relations, due to their major economic impact and their increasing use in contemporary geopolitical conflicts.

Cyberattacks as a domain of interstate conflict A cyberattack can be defined as any intentional act using information technology to compromise the availability, integrity, or confidentiality of systems or data, resulting in significant disruption or theft of information (Iftikhar, 2024). Beyond this technical dimension, several authors in international relations emphasize that cyberspace has become a new domain of conflict, on top of land, sea, air, and space. Institutional work, particularly that of NATO (2016), explicitly recognizes cyberspace as a military area of operations, confirming that cyberattacks are now part of conflict strategies and geopolitical influence. Rid and Buchanan (2015) further show that cyber operations offer several strategic advantages over conventional weapons: relatively low entry costs, attribution difficulties, and the ability to produce significant economic or societal damage without formally crossing the threshold of armed conflict. This ambiguity complicates diplomatic and military responses and reinforces international instability. Valeriano and Maness (2014) provide empirical evidence on this point, documenting that cyber conflict between rival state dyads is persistent, escalatory, and closely tied to pre-existing geopolitical rivalries, suggesting that cyber and physical hostilities are strategically intertwined rather than independent phenomena.

Strategic and economic consequences of cyberattacks The growing use of cyberattacks is illustrated by several emblematic cases. The WannaCry ransomware campaign of 2017 and the

ransomware attack against Colonial Pipeline in 2021 demonstrate the capacity of cyber operations to disrupt critical infrastructure for purposes of political pressure, economic destabilization, or support to conventional military operations ([Aljohani, 2022](#)).

The economic consequences of such attacks extend well beyond the directly targeted organizations. According to [World Economic Forum \(2023\)](#), cyberattacks rank among the major global risks due to their capacity to disrupt global value chains and undermine economic stability. Direct financial losses, high recovery costs, increased cybersecurity expenditure, and supply chain disruptions can generate negative externalities affecting entire national economies. In a tense geopolitical context, these attacks therefore function as instruments of indirect pressure, allowing states to weaken an adversary's economy without resorting to formal sanctions or direct military intervention.

Cyberattacks and hybrid warfare Hybrid warfare is characterized by the simultaneous combination of conventional and unconventional instruments of power, including disinformation, economic coercion, clandestine operations, and cyber activity ([Iftikhar, 2024](#)). In this context, cyberattacks serve multiple strategic functions: preparing the operational ground before a conventional military engagement, probing an adversary's response capabilities, or maintaining sustained pressure below the threshold of declared war. The attacks on Ukrainian power infrastructure illustrate this logic, combining kinetic and cyber pressure in a coordinated manner alongside conventional military pressure ([Aljohani, 2022](#)). From this perspective, cyberattacks are now vectors for transforming contemporary conflict, reshaping the balance of power and altering traditional mechanisms.

2.3 Economic literature

A large literature documents the negative economic consequences of armed conflicts, and shows that poverty and conflict tend to reinforce each other. The economic impact of cyberattacks has received less attention and remains an emerging area of research.

Conflict trap Economic development can play a crucial role in conflicts and international relations between countries. As countries develop, they tend to have a bigger army as well as larger economic and political power. On the other side, under-developed countries seek to improve growth, international trade, and decision power but they can face a major problem: civil wars.

[Nurkse \(1953\)](#) in his book “Problems of Capital Formation in Underdeveloped Countries” is one of the first to materialize the notion of poverty trap, or vicious circle of poverty. The idea is that poor countries with low income have low savings which implies low investment which itself implies low productivity. This “circular constellation of forces tending to act and react upon one another in such a way as to keep a poor country in a state of poverty” ([Nurkse, 1953](#)). Later on, based on the same idea of vicious circle, appears the notion of conflict trap. “Breaking the conflict trap”, from [Collier et al. \(2003\)](#) remains the fundamental reference on it. Indeed, this world bank report argues that wars are a major obstacle to macroeconomic development. More particularly, they focus on civil wars as they are more frequent than interstate conflict and try to estimate a causal effect on poverty and conflict. The main results underline that under development is a causal and persistent significant effect of civil war. Intrastate conflict can be retarded by development, but, if there is a conflict, the development will be retarded. In addition, conflicts are closely related to macroeconomic variables (GDP, population). In “Does poverty cause conflict? Isolating the causal origins of the conflict trap”, [Braithwaite et al. \(2016\)](#) find similar results by estimating the causal effect of poverty on conflict. The main results show that again poverty, measured by GDP per capita and infant mortality, has a strong significant impact on intrastate conflict as well as population size whereas temporal dependence, measured by the number of peace years before a conflict, is not significant.

According to existing literature, once a country experiences conflict, it faces a negative impact on its economic development which tends to increase the probability of a new conflict. Poverty can be seen as the main cause of this effect.

International trade Macroeconomics indicators have a significant impact on intrastate conflict, nevertheless, we are interested in studying both intrastate and interstate conflicts. Is there a causal effect between economic indicators and interdependent wars ? When thinking about economic indicators between two countries we firstly think of international trade. Focusing on the latter [Gartzke et al. \(2001\)](#) and [Martin et al. \(2008\)](#), respectively in “Investing in the Peace: Economic Interdependence and International Conflict” and “Make trade not war ?” highlight the negative significant impact of international trade on interstate conflicts. Indeed, due to trade, the opportunity cost of conflict and the economic impact of such an event are much higher than for independent states.

Trade can be seen as trade of goods but it could also imply monetary and financial interdependence. [Martin et al. \(2008\)](#) highlight the difference between bilateral dependence and global dependence. Bilateral trade increases the cost of opportunity for each of the two countries, nevertheless, global trade decreases the bilateral dependence as well as the opportunity cost of war. To wrap up, they state that bilateral dependence decreases much more the probability of conflict than global trade.

2.4 Hypothesis development

Hypothesis 1 – Bidirectional dynamics between physical conflicts and cyberattacks

The existing literature identifies contiguity, power parity, bilateral trade, democracy, and membership in international organizations as the main determinants of interstate conflict. However, none of these frameworks explicitly accounts for the role of cyber operations. As cyberspace has progressively become a recognized domain of military confrontation, the emergence of hybrid warfare strategies suggests that cyberattacks now constitute both a complement to and a potential trigger of physical violence. The growing share of cyber incidents in global risk assessments, combined with documented cases of cyber operations preceding or accompanying conventional military engagements, supports the hypothesis that the relationship between cyber activity and physical conflict is not unidirectional. **We therefore hypothesize that physical conflicts trigger cyber escalation, and that cyberattacks in turn increase the probability of physical violence,** forming a mutually reinforcing feedback loop that we formally test in the first part of our empirical analysis.

Hypothesis 2 – Economic consequences of conflicts and cyberattacks The negative impact of armed conflict on economic development is well established in the literature: wars represent a major obstacle to macroeconomic growth, with persistent effects on GDP per capita, investment, and human capital accumulation. If, as suggested by the first hypothesis, physical conflicts and cyber operations are complementary and mutually reinforcing, then cyberattacks should also exert a negative effect on economic performance. **We therefore hypothesize that both armed conflicts and cyberattacks have a negative and significant impact on GDP per capita,** and we assess this question quantitatively in the second part of our empirical analysis.

Hypothesis 3 – Dyadic predictors of conflict onset and persistence At the dyadic level, the literature shows that bilateral trade and economic interdependence significantly reduce the probability of interstate conflict by raising the opportunity cost of war. To date, however, the predictive role of cyber incidents in conflict dynamics has received little empirical attention. Given the evidence on cyber-conflict complementarity documented in the first hypothesis, cyber activity between two countries may carry important information about the likelihood of physical confrontation. **We therefore hypothesize that cyber incidents between two countries are a significant predictor of both conflict onset and conflict persistence**, a claim we assess using rare event logit models and non-parametric machine learning approaches in the third part of our empirical analysis.

3 Data

In this paper we rely on different datasets. The following section presents each of them and how they have been used in this paper.

3.1 Data Sources

The version 25.1 of the Dyadic Dataset ([Harbom et al., 2008](#); [Davies et al., 2025](#); [Pettersson, 2025](#)) from the *Uppsala Conflict Data Program (UCDP)*, at the *Department of Peace and Conflict Research, Uppsala University* reports 3,432 armed conflicts by dyad of countries from 1946 to 2024. According to the UCDP, a dyadic armed conflict involves two parties of which at least one is the government of a state, and such that there are at least 25 battle-related deaths in a year. This dataset contains 25 variables such as the two sides of the conflict, the start and end dates, the reason of incompatibility between the two parties, the type of conflict, its intensity, its outcome and settlement, but also the role of each state in the dispute along with the highest military confrontation action in the dyadic dispute. From this dataset, we extract all the conflicts that involve at least one recognized state.

The version 1.3 of the *Global Dataset of Cyber Incidents from the European Repository of Cyber Incidents* ([EuRepoC, 2023](#)) contains data on 3,414 cyber incidents between January 1, 2000 and December 31, 2024. The repository only includes cyber incidents that affected the Confidentiality,

Integrity or Availability of the targeted systems. This dataset counts 85 variables, the main ones being the incident type, the countries involved in the attack (receiver and initiator), the types of parties involved (state-affiliated actor, non-state group, etc.), and the economic impact of the attack. As this dataset only covers the period from 2000 onwards, the analysis in this paper is restricted to 2000–2024 to ensure time consistency across all sources.

In addition to these two main datasets, several complementary sources are used to enrich the analysis with macro, trade, and dyadic variables: the CEPII Gravity database ([Conte et al., 2023](#)), the International Trade in Goods by Partner Country from the [IMF \(2025\)](#), the Polity5 dataset ([Marshall & Gurr, 2020](#)), the version 5.1 of the Alliance Treaty Obligations and Provisions (ATOP) dataset ([Leeds et al., 2002](#)), the version 6.0 of the National Material Capabilities dataset from the Correlates of War (NMC–COW) project ([Singer et al., 1972](#)), and the World Development Indicators ([WorldBank, 2024](#)).

3.2 Dataset Construction

From these databases we build the two main datasets used in this paper. The first is a dyadic panel dataset at the country-pair-year level, covering the period 2000 to 2024. Conflict and cyber incident data are first aggregated separately before being merged into a unified dyadic structure. Armed conflict exposure is measured by counting, for each country and each year, the number of active conflicts in which the country is involved. This count is then assigned to each dyad by adding the conflict exposure of the two countries in the pair. Cyber incidents are initially recorded with an identified attacker and an identified target country. Since we do not distinguish between the direction of the attack, each incident is counted as an interaction between the two countries involved, regardless of which one initiated it. Alliance ties are captured through a binary variable equal to one if the two countries have at least one alliance agreement in a given year. The universe of dyads is then restricted to politically relevant pairs, defined as pairs that experienced at least one conflict or one cyber incident over the sample period, which reduces the size of the panel while preserving the variation of interest. Macro, trade and dyadic covariates are subsequently merged onto this structure using ISO3 and Correlates of War country codes as matching keys, depending on the source. These covariates include, for each country in the pair, GDP per capita, total population,

the share of the population using the internet, mobile subscriptions, the Polity2 democratic score, and the CINC military capability index, as well as dyadic variables such as bilateral distance, contiguity, common official language, common religion, colonial ties, trade agreement membership, GATT/WTO and EU membership, and bilateral trade flows. Variables with more than 30% of missing values are excluded from the final dataset, resulting in a panel of 13,856 country-pair-year observations, hereafter referred to as **DyadPanel**.

The second dataset is a country-level panel derived from the dyadic one, obtained by stacking both sides of each dyad and aggregating conflict and cyber exposure at the country-year level, yielding 3,800 country-year observations, referred to as **CountryPanel** hereafter. **CountryPanel** includes GDP per capita, total population, the share of the population using the internet, mobile subscriptions, and the number of conflicts and cyber incidents per country, per year.

3.3 Descriptive statistics

The descriptive statistics of the numeric variables of **CountryPanel** and **DyadPanel** are respectively reported in [Table 1](#) and [Table 2](#).

[Figure 1](#) illustrates the trend of the two series: conflicts and cyberattacks. Since 2000, both series follow a similar increasing trend with a huge peak in 2022 for cyberattacks and 2023 for conflicts. These peaks are followed by a sharp fall in the number of events the year after.

By comparing the dyads, [Figure 4](#) shows the Top 20 in number of incidents and armed conflicts for the 2000-2024 period. Concerning cyber events, the three most important dyads are Russia-Ukraine (398 incidents), China-USA (346) and Russia-USA (196). For armed conflicts, the top pairs are India-Pakistan (211), China-India (65) and Afghanistan-Pakistan (46). By combining both type of events, [Figure 5](#) illustrates the most actives dyads in the datasets: Russia-Ukraine in first, then India-Pakistan and China-USA.

By taking the percentage of dyads-years with onset and the number of years with peace (i.e. without conflict), [Figure 8](#) shows that when the peace duration (in years) is larger, the percentage of dyads with a new conflict is higher.

From the ongoing war point of view, [Figure 9](#) illustrates the fact that dyads with 6 years of ongoing conflicts are the most likely to have persistence in their conflict. By contrast, with only one year

of conflict there are less than 25% of the dyad-years with persistence.

Lastly, the network graph ([Figure 10](#)) gathers physical conflicts and cyberattacks for dyads of countries. There are strong tensions between China and USA, dominated by cyberattacks. The purple lines between Iran-Israel, India-Pakistan or USA-Iran highlight a dominance of physical conflicts. The pink link for Russia-Ukraine shows an approximate balance between physical and digital disputes.

4 Empirical strategy

This paper addresses three empirical questions that flow naturally from one another. The starting point lies in establishing that cyber and physical conflicts are mutually reinforcing. Second, we assess whether this escalation dynamic translates into measurable economic costs. Third, we investigate whether cyber activity helps to predict which country pairs are most likely to experience or sustain conflict. Each question requires a dedicated methodological framework, described below.

Do physical conflicts trigger cyber operations, and conversely, do cyberattacks escalate into physical violence? To answer this first problematic, we rely on a Panel VAR estimated via system GMM that capture short-run cross-country dynamics. This is complemented by a VECM and an aggregate VAR to characterize the long-run equilibrium relationship and global shock transmission at the aggregate level.

In order to examine the economic consequences of both types of hostilities, we use two-way fixed effects panel models. This is done by following the framework of [Vergara Cobos et al. \(2024\)](#), which we extend by jointly including physical conflict exposure.

Finally, we investigate the dyadic determinants of physical conflicts at the country-pair level: what characteristics make two countries more likely to initiate a physical conflict, and conditional on a conflict having already occurred, what factors determine whether it persists over time? A particular focus is placed on the role of cyber incidents. Given the binary and rare nature of physical conflicts at the dyadic level, we rely on a rare events logit model and complement it with non-parametric machine learning approaches, namely Random Forest and Extreme Gradient Boosting, to capture potential non-linearities and interactions that a linear model would miss.

The first two questions concerning bidirectional dynamics and economic impact, are addressed using the **CountryPanel** dataset. The third question, which requires information on country-pair characteristics, is addressed using the **DyadPanel** dataset.

5 Bidirectional dynamics between conflicts and cyberattacks

5.1 Models Specification

Panel VAR The standard reduced-form vector auto regressive (RF-VAR) framework assumes homogeneous dynamics across observational units. In a cross-country setting, however, each country i possesses time-invariant structural characteristics, denoted μ_i , which are likely correlated with the endogenous variables. Ignoring these country-specific effects generates omitted-variable bias. Moreover, the inclusion of lagged dependent variables induces dynamic endogeneity, making fixed-effects OLS inconsistent in short panels, this effect is referred as the Nickell bias (Nickell, 1981). Therefore, an appropriate framework must control for unobserved heterogeneity, correct for endogeneity induced by lagged dependent variables and preserve the multivariate dynamic structure of the VAR. This motivates the use of a dynamic Panel VAR estimated via system GMM. Indeed, the Panel VAR isolates short-run country-level dynamics while controlling for structural heterogeneity, and allows panel Granger causality testing. Extending the RF-VAR(p) to the panel dimension, the specification of the system is:

$$Conflict_{it} = \alpha_{11} Conflict_{i,t-1} + \alpha_{12} Cyber_{i,t-1} + \mu_i^{Conflict} + \varepsilon_{it}^{Conflict} \quad (1)$$

$$Cyber_{it} = \alpha_{21} Conflict_{i,t-1} + \alpha_{22} Cyber_{i,t-1} + \mu_i^{Cyber} + \varepsilon_{it}^{Cyber} \quad (2)$$

Where:

$Conflict_{it}$ is the number of armed conflicts in country i and year t ,

$Cyber_{it}$ is the number of cyberattacks in country i and year t ,

$\mu_i^{Conflict}$ and μ_i^{Cyber} are country fixed effects,

$\varepsilon_{it}^{Conflict}$ and ε_{it}^{Cyber} are innovations.

To remove country fixed effects, we apply a forward orthogonal deviations (FOD) transformation. The system GMM estimator of Blundell and Bond (1998) then uses lagged levels as instruments for

the transformed equations and lagged differences as instruments for the levels equations. To avoid instrument proliferation, the instrument matrix is collapsed following [Roodman \(2009\)](#). We use lags 3 of the dependent variables as instruments. The Hansen test of overidentifying restrictions ($\chi^2(8) = 12.32$, $p = 0.138$) does not reject the null hypothesis of instrument validity, confirming that the instrumentation strategy is appropriate and that the moment conditions are not systematically violated.

Having characterized short-run dynamics at the country level with the Panel VAR, we therefore move to the analysis of long-run equilibrium relationships at the global level. The next step is then to examine the properties of the aggregated series across countries to determine their integration and cointegration patterns, which will guide the specification of the aggregate long-run model.

Integration and Cointegration Testing Before specifying the global dynamic model, we assess the order of integration of the series to avoid spurious regressions ([Granger and Newbold, 1974](#)). Augmented Dickey-Fuller tests suggest that both Conflict and Cyber are non-stationary in levels. While Conflict is clearly stationary in first differences, results for Cyber are less conclusive. Complementary KPSS tests show stationarity in first differences, leading us to treat both series as $I(1)$. Lag selection criteria suggest a $VAR(2)$, but cointegration is not detected.

Given this limitation, we impose $r = 1$ cointegrating relationship. This assumption is supported by the empirical literature, which documents a common upward trend in both armed conflicts and cyberattacks driven by global geopolitical dynamics ([Themnér and Wallenstein \(2014\)](#) and [Valeriano and Maness \(2014\)](#)). The use of a VECM is then only justified on economic grounds. As a consequence, all VECM results should be interpreted with this caveat in mind: the long-run relationship is assumed rather than statistically confirmed, the estimated parameters may be less precise than in a large-sample setting, and the VECM should not be interpreted as a fully data-driven equilibrium representation.

Vector Error Correction Model (VECM) With 2 lags, the specification is the following:

$$\Delta Conflict_t = \alpha_{Conflict} \underbrace{(\beta_1 Conflict_{t-1} + \beta_2 Cyber_{t-1})}_{ECT_{t-1}} + \Gamma_{Conflict1} \Delta Y_{t-1} + \varepsilon_t^{Conflict} \quad (3)$$

$$\Delta Cyber_t = \alpha_{Cyber} ECT_{t-1} + \Gamma_{Cyber1} \Delta Y_{t-1} + \varepsilon_t^{Cyber} \quad (4)$$

Where:

$\Delta Conflict_t$ represents the annual changes in number of conflicts,

$\Delta Cyber_t$ represents the annual changes in number of cyberattacks,

$ECT_{t-1} = \beta_1 Conflict_{t-1} + \beta_2 Cyber_{t-1}$ is the error-correction term capturing the long-run disequilibrium.

$\alpha_{Conflict}$ measures the speed at which the conflict variable adjusts back to the long-run equilibrium,

α_{Cyber} measures the speed at which the cyber variable adjusts back to the long-run equilibrium,

$\Gamma_{Conflict1}$ and Γ_{Cyber1} are 1×2 coefficient vectors on the first lag of the differences,

$\Delta Y_{t-1} = (\Delta Conflict_{t-1}, \Delta Cyber_{t-1})$ captures short-run interactions between conflicts and cyberattacks,

$\varepsilon_t^{Conflict}$ and ε_t^{Cyber} denote the innovations or residuals.

Aggregate VAR While the VECM identifies long-run equilibrium relationships, it does not directly provide structural impulse responses (IRFs), variance decompositions (FEVD), or forecasting tools in their standard VAR form. To analyze systemic shock transmission and global dynamic interactions, we rely on the VAR representation implied by the VECM, which preserves the long-run cointegration constraints while allowing for a short-run structural analysis.

5.2 Results

Panel VAR [Table 6](#) presents the results of the Panel Var estimation. In the Conflicts equation, past conflicts are statistically significant, suggesting a detectable short-term persistence of conflict on the next period but not direct effect of cyber activity on physical conflicts at the one-year lag.

In the Cyberattacks equation, both past cyberattacks and conflicts are highly significant, indicating strong autocorrelation: cyber operations tend to persist from year to year. The effect of past conflicts imply that conditional on the model, there is a clear short-term escalation from conflicts to cyberattacks. This pattern is consistent with the increasing use of cyber operations as a

complementary tool of modern war, where states and non-state actors extend traditional conflicts into the digital space. Cyberattacks may therefore be interpreted as a strategic continuation of physical hostilities.

Overall, the Panel VAR results point to a clear asymmetry: cyberattacks and physical conflicts exhibit strong internal dynamics and self-reinforcing persistence, while we identify a clear escalation of cyber conflicts after physical conflicts. This finding motivates the global-level analysis, where aggregate dynamics and longer run equilibrium relationships may reveal patterns that are not visible in the cross-country short-run variation.

Vector Error Correction Model (VECM) Despite the weak statistical evidence for cointegration, the VECM provides a theoretically grounded framework to study both the long-run equilibrium between conflicts and cyberattacks and the short-run adjustment dynamics. The estimated cointegration vector is:

$$ECT_t = Conflict_t - 0.251 Cyber_t - 11.639. \quad (5)$$

This implies the long-run relationship ($ECT_{t-1} = 0$ in the long run):

$$Conflict_t \approx 0.251 Cyber_t + 11.639. \quad (6)$$

Economically, this indicates a strong long-term association: one additional cyberattack in year t corresponds to an increase of approximately 0.251 physical conflicts in the long run. This positive long-run relationship reflects a complementarity and escalation effect: as armed conflicts intensify, state and non-state actors simultaneously escalate using the cyber arm, suggesting that conflict and cyber activity are strategic complements rather than substitutes.

The estimated short-run adjustment equations are:

$$\Delta Conflict_t = -0.334 \cdot ECT_{t-1} - 0.459 \cdot \Delta Conflict_{t-1} + 0.107 \cdot \Delta Cyber_{t-1} \quad (7)$$

$$\Delta Cyber_t = 5.343 \cdot ECT_{t-1} + 5.681 \cdot \Delta Conflict_{t-1} - 0.696 \cdot \Delta Cyber_{t-1} \quad (8)$$

The error-correction terms reveal an asymmetry in how the two variables respond to long-run disequilibria, and constitute one of the most important results of this paper. Since $\alpha_{Conflict} = -0.334 < 0$, the conflict series is the error-correcting variable: whenever the system drifts away from its long-run equilibrium, it is conflicts that adjust to restore the balance. This implies that physical conflicts act as a stabilizing force in the system. By contrast, $\alpha_{Cyber} = 5.343 > 0$ indicates that cyberattacks move away from equilibrium in response to disequilibrium, amplifying rather than correcting deviations from the long-run path. Cyber dynamics therefore act as a source of systemic instability. This asymmetry has an important economic interpretation: even if the long-run relationship between conflicts and cyberattacks is positive and stable, it is maintained by the conflict variable alone.

In the short run, conflicts precede cyberattacks ($5.681 > 0$, $p < 0.01$), consistent with the Panel VAR finding that armed conflicts Granger-cause cyber activity. The coefficient on $\Delta Cyber_{t-1}$ in the conflict equation (0.107) is positive, suggesting a potential feedback from cyber to conflict in the short run. Together, these dynamics point to a feedback loop in which conflicts and cyberattacks mutually reinforce each other in the short run. Finally, the negative own-lag coefficients (-0.459 and -0.696) reflect mean-reverting behavior in both series, indicating that short-run deviations in each variable tend to correct themselves over time.

Aggregate VAR Before developing on the interpretation of the results, the validity of the model is checked through a series of diagnostic tests. First, all roots of the characteristic polynomial lie strictly inside the unit circle, confirming the stability of the estimated VAR(2). Second, the Portmanteau test for residual autocorrelation does not reject the null hypothesis of no serial correlation, indicating that the lag structure adequately captures the dynamics of the data. These diagnostics statistically confirm the validity of the VAR specification.

The aggregate VAR is estimated on 23 annual observations (2002–2024) and results are reported in [Table 7](#). As it is standard in VAR analysis, individual coefficients are not directly interpretable due to multicollinearity across lags and the simultaneous feedback structure of the system. We therefore focus on Granger causality tests, impulse response functions, and forecast error variance

decompositions to draw conclusions.

Granger causality These tests confirm bidirectional predictability. Indeed past cyberattacks significantly Granger cause conflicts ($p = 0.0002$), and past conflicts significantly Granger cause cyberattacks ($p = 0.008$), suggesting that conflicts and cyberattacks are mutually reinforcing at the global level in the short run.

Impulse response function To better understand the dynamic interactions identified above, we now turn to impulse response analysis. [Figure 11](#) reports the orthogonalized Impulse Response Functions, with a Choleski ordering placing conflicts first. This ordering choice is consistent with the weak exogeneity of conflicts identified in the VECM, which implies that conflicts do not immediately adjust to short-run disequilibria and are therefore treated as the most exogenous variable in the system. The bands represent 95% confidence intervals.

Conflicts $\rightarrow$ Conflicts. A one standard deviation shock to conflicts generates a large, positive and persistent response in conflicts themselves, starting at approximately 50 at impact and oscillating between 25 and 70 over the ten-year horizon. The confidence band remains above zero for almost all the period, indicating a statistically significant and permanent effect. This oscillatory pattern reflects the VAR(2) dynamics identified in [Table 7](#), and is consistent with the VECM result that conflicts act as the error-correcting variable: shocks to conflicts are not fully absorbed in the short run but cumulate and propagate over time.

Conflicts $\rightarrow$ Cyberattacks. A shock to conflicts generates an immediate negative response in cyberattacks at impact (approximately -150), followed by large oscillations between -400 and $+30$ over the horizon. The confidence band crosses zero throughout, meaning the effect is not statistically significant at conventional levels. The sign alternation is consistent with the negative coefficient on the second lag of conflict identified in the cyberattack equation (-7.907 , $p = 0.004$): a conflict shock initially displaces cyber activity before triggering a delayed escalation response.

Cyberattacks $\rightarrow$ Conflicts. The response is zero at impact by construction of the Cholesky ordering, then turns positive and significant from horizon 1, reaching approximately 40 conflicts at horizon 2, before oscillating and returning toward zero at horizon 10. The confidence band excludes zero at horizons 1,2 and 6,7, providing partial statistical support for the Granger causality

result (Cyber $\rightarrow$ Conflicts, $p = 0.0002$). This pattern is also consistent with the VECM short-run dynamics: the positive loading $\alpha_{Cyber} = 5.343$ indicates that cyberattacks amplify disequilibria rather than correcting them. This observation suggests that cyber shocks can gradually escalate into physical conflicts over a two-year horizon.

Cyberattacks $\rightarrow$ Cyberattacks. A shock to cyberattacks generates a large immediate response (approximately 450 units at impact), which oscillates but remains globally positive over the horizon. The confidence band excludes zero at most horizons. This strong persistence is consistent with the VECM finding that $\alpha_{Cyber} = 5.343 > 0$: once initiated, cyber campaigns sustain and amplify themselves over multiple periods rather than mean-reverting.

Forecast error variance decomposition To complement the IRF analysis, we examine the forecast error variance decomposition. [Figure 12](#) and [Figure 13](#) report them, decomposing the forecast error variance of each variable into contributions from conflict shocks and cyber shocks at horizons 1 to 10.

FEVD: Conflicts. At horizon 1, 100% of the forecast error variance in conflicts is explained by its own shock, which is mechanical, given the Cholesky ordering, placing conflicts first. At horizon 2, cyber shocks, account for approximately 40% of conflict forecast error variance, a share that grow gradually, reaching approximately 50% by horizon 10. Over the medium and long run, cyber shocks account for roughly half of the unexpected variation in conflicts, placing cyber activity on an equal place with conflict's own dynamics as a driver of physical violence. This is consistent with the Granger causality result (Cyber $\rightarrow$ Conflicts, $p = 0.0002$) and reinforces the interpretation drawn from the IRF analysis: cyber operations exert a persistent and growing influence on physical conflicts over time, suggesting again that the two phenomena are deeply intertwined rather than evolving independently.

FEVD: Cyberattacks. At horizon 1, conflict shocks account for approximately 6% of the forecast error variance in cyberattacks, with the remaining 94% explained by cyber's own dynamics. The contribution of conflict shocks increases rapidly, reaching approximately 16% at horizon 2 and stabilizing around 30% from horizon 3 onwards. While cyberattack dynamics remain predominantly self-driven across all horizons, the non-negligible and growing share attributed to conflict shocks is consistent with the Granger causality result (Conflicts $\rightarrow$ Cyberattacks, $p = 0.008$) and

with the VECM long-run relationship: physical conflicts exert a meaningful influence on cyber activity over the medium run, even if cyber campaigns also follow a partly autonomous logic driven by technological and strategic factors.

Overall, the IRF and FEVD analyses deliver a consistent and mutually reinforcing picture. Both series exhibit strong own persistence, confirming the feedback loop identified through Granger causality testing. More importantly, the cross effects reveal an asymmetric but bidirectional relationship: cyber shocks account for a growing share of conflict forecast error variance, reaching approximately 50% at horizon 10, while conflict shocks explain around 30% of cyber variance from horizon 3 onwards. This asymmetry is consistent across all three empirical frameworks employed in this section (the Panel VAR, the VECM, and the aggregate VAR) and points to a complex escalation mechanism in which physical and cyber conflicts are deeply intertwined. Cyber operations increasingly shape the trajectory of armed violence in the medium and long run, while conflicts in turn sustain and amplify cyber activity.

5.3 Take away

The results of this first section provide strong support for Hypothesis 1, highlighting the existence of self reinforcing dynamics between cyberattacks and physical conflicts. More precisely, physical conflicts appear as the starting point of the escalation process: they trigger cyber activity in the short run, while cyberattacks in turn amplify and sustain physical violence over time. In the long run, this feedback loop is asymmetric. Indeed, conflicts act as the stabilizing variable that corrects deviations from equilibrium, while cyber activity is the destabilizing force that pushes the system away from it. If cyber and physical conflicts indeed reinforce each other as shown here, one would expect this dynamic to carry economic costs. The next section is about assessing this possible impact.

6 Economic consequences of conflicts and cyberattacks

Having established the dynamic interactions between conflicts and cyberattacks, we now turn to their economic consequences.

6.1 Models Specification

Two Way Fixed Effect Panel Models The motivation for these models comes directly from [Vergara Cobos et al. \(2024\)](#). In order to determine the economic impact of cyberincidents on economies, these authors consider two main specifications, with and without control variables:

$$\log(GDP_{pcit}) = \beta_1 \log(Cyber_{it}) + \alpha_i + \lambda_t + \varepsilon_{it} \quad (9)$$

$$\begin{aligned} \log(GDP_{pcit}) = & \beta_1 \log(Cyber_{it}) + \beta_2 Internet_{it-3} \\ & + \beta_3 \log(Mobile_{it-3}) + \beta_4 \log(Pop_{it-3}) \\ & + \alpha_i + \lambda_t + \varepsilon_{it} \end{aligned} \quad (10)$$

We extend this framework by adding the logarithm of the number of physical conflicts in order to answer completely our problematic.

$$\log(GDP_{pcit}) = \beta_1 \log(Cyber_{it}) + \beta_2 \log(Conflict_{it}) + \alpha_i + \lambda_t + \varepsilon_{it} \quad (11)$$

$$\begin{aligned} \log(GDP_{pcit}) = & \beta_1 \log(Cyber_{it}) + \beta_2 \log(Conflict_{it}) \\ & + \beta_3 Internet_{it-3} + \beta_4 \log(Mobile_{it-3}) \\ & + \beta_5 \log(Pop_{it-3}) + \alpha_i + \lambda_t + \varepsilon_{it} \end{aligned} \quad (12)$$

Where:

GDP_{pcit} is the GDP per capita in country i and year t ,

$Conflict_{it}$ is the number of armed conflicts in country i and year t ,

$Cyber_{it}$ is the number of cyberattacks in country i and year t ,

$Internet_{it}$ is the percentage of internet users in country i and year t ,

$Mobile_{it}$ is the total number of mobile subscriptions in country i and year t ,

Pop_{it} is the population size in country i and year t .

For the control variables, as in [Vergara Cobos et al. \(2024\)](#) we only include the three-period lagged values to minimize the direct relationship between the control variables and the treatment variable.

6.2 Results

The four models estimate the impact of $\log_Cyber$ on GDP per capita within a two-way fixed effects framework (country and year fixed effects).

Without controlling for physical conflict (Table 8 and Table 9), the coefficient on $\log_Cyber_{it}$ is not statistically significant, regardless of whether technological control variables are included. However, once $\log_Conflict_{it}$ is introduced (Table 10 and Table 11), the coefficient on $\log_Cyber_{it}$ becomes significant at the 5% level. This suggests that, in the absence of this control, we had an omitted variable bias. Indeed the cyber variable was partially capturing the effect of physical conflicts, with which it is positively correlated.

The positive sign on $\log_Cyber_{it}$ (ranging from +0.013 to +0.018) is counter-intuitive at first glance. Rather than reflecting a direct positive effect of cyberattacks on economic output, this result likely reflects an endogeneity problem. Addressing this issue would require an instrumental variable approach to account for simultaneity and reverse causality. This is left for a future work.

Finally, the coefficient on $\log_Conflict_{it}$ is never statistically significant across specifications ($p \approx 0.21\text{--}0.25$), meaning that we cannot identify a robust direct effect of physical conflicts on GDP per capita in this framework. This null result may reflect measurement limitations: armed conflicts tend to generate highly localized economic disruptions that are difficult to detect when GDP is measured at the national level.

6.3 Take away

The results of this section provide limited evidence supporting Hypothesis 2. This is likely due to identification challenges. Indeed the relationship between cyberattacks, conflicts and economic performance is more complex than a simple causal link, and addressing it rigorously would require an instrumental variable strategy. These challenges also motivate the shift to a dyadic perspective: rather than estimating aggregate economic impacts, we now investigate whether cyber and economic variables help to predict which specific country pairs are most at risk of physical confrontation, using models that account for the rare nature of conflict events and that impose no functional form on the covariates.

7 Dyadic determinants of cyber incidents

While the previous sections focused on aggregate and country-level dynamics, we now shift to a dyadic perspective to better understand the determinants of conflict between specific country pairs.

7.1 Models Specification

The purpose of this last section is to identify the main determinants of conflicts between two countries in order to perform out-of-sample forecasts on their probability of occurrence. Given the current geopolitical situation, the final goal of this section (and this paper) is to do forecasting on the probability of armed conflict for the following pairs:

- **Russia-Ukraine:** Active war since 2022, most documented cyber-physical conflict of the decade.
- **Iran-Israel:** Direct military exchanges in 2024, mutual declared hostility, nuclear dimension.
- **Iran-USA:** Decades of proxy confrontation, sanctions regime, direct strikes in early 2026.
- **India-Pakistan:** Nuclear dyad, Kashmir dispute, armed confrontation in May 2025.
- **Russia-USA:** NATO escalation risk, nuclear deterrence, proxy confrontation in Ukraine.
- **Afghanistan-Pakistan:** Shared border, Taliban spillovers, recurring cross-border incidents and armed clashes in October 2025.

Note that we focus on these dyads only for the forecasting part of the paper. In order to identify the main determinants of conflicts and evaluate performances we use the entire dyad-year sample.

Conflict onset and persistence A distinction is done between the dyads that are already in conflict and the dyads that are not. Indeed, it is different to predict conflict onset (i.e. new conflict) and persistence of a current dispute. The `DyadPanel` dataset is then divided into two other datasets. The `Onset` one, contains 11,092 observations for which, considering a given dyad-year observation, no conflict is observed the preceding year ($lag_conflict = 0$). Similarly, the `Persist` dataset contains 668 observations for which a conflict is identified in the previous year.

The models presented in this section are estimated two times, one for each of these two datasets with respectively *conflict_onset* and *conflict_persistence* as dependent variables.

Sample splitting For **Persist** and **Onset**, the data is separated into training, validation and testing sets, in order to evaluate the models performances. The training sets are from 2000 to 2019 (included). The validation includes years from 2015 to 2019, and the testing set is from 2020 to 2024 (included). In addition, out of sample forecasts are performed on the selected dyads for 2025 which is not observed in the available data.

Rare events Logit The first logit model estimated is used to predict the probability of conflict onset in a dyad i at year t , conditional on the fact that there was no conflict in the previous year. The second logistic model is used to estimate the probability of a conflict persistence in a dyad i in year t , conditional on the fact that there was already a conflict in the previous year. Respectively:

$$\text{logit}(P(Y_{it}^{Onset} = 1)) = \alpha + X_{it}^{Onset}\beta + \gamma_t \quad (13)$$

$$\text{logit}(P(Y_{it}^{Persistence} = 1)) = \alpha + X_{it}^{Persistence}\beta + \gamma_t \quad (14)$$

With X_{it} the vector of covariates. It includes conflict history, cyber history, political, economical and gravity variables. For the conflict onset model, the covariates are listed in [Table 3](#), while for the persistence model they are listed in [Table 4](#). γ_t is year fixed effects.

Based on [King and Zeng \(2001\)](#), the logit model is biased with rare events. Indeed, when the dependent variable has many more zeros than ones, using a simple logistic regression "can sharply underestimate the probability of rare events" ([King and Zeng , 2001](#)). The MLE estimator $\hat{\beta}$ obtained by maximizing the log-likelihood may suffer from small-sample bias. This issue is particularly concerning in our dataset, where the proportion of conflict observations is very low. In the **Persist** dataset, there are 31.4% of conflicts when there are 4.6% in the **Onset** one ¹. This justifies the need for bias correction. To address this, we use the bias-reduction of [Firth \(1993\)](#) which introduces a penalized MLE. Instead of maximizing the standard log-likelihood ($\ell(\beta)$), the

¹For the entire sample period (2000-2024)

estimator maximizes the penalized log-likelihood:

$$\ell^*(\beta) = \ell(\beta) + \frac{1}{2} \log |I(\beta)|,$$

where $I(\beta)$ denotes the Fisher information matrix.

The resulting $\hat{\beta}^{Firth}$ estimator removes the first-order bias of the MLE.

Non Parametric Models (Random Forest and Extreme Gradient Boosting) The logistic model imposes strong assumptions, in particular linearity in the covariates. However, the relationship between cyber threats, economic factors, and conflict is likely to be non-linear and to involve complex interactions. To relax the linearity assumption and capture richer patterns in the data, we complement our analysis with non-parametric models: Random Forest and Extreme Gradient Boosting.

Random forest is an ensemble method using bagging. This technique leads to lower variance and is robust against overfitting.

Extreme Gradient Boosting (XGBoost) is based on the boosting method and minimizes the error using gradient descent. This algorithm is particularly known for its high accuracy.

Ensemble model In addition, we construct an *Ensemble* model which is a combination of the rare event logit, random forest, and XGBoost out-of-sample predictions through a weighted average. The weights are selected by grid search on the validation set so as to maximize the AUC:

$$\hat{\mathbb{P}}_i^{\text{Ensemble}} = w_1 \hat{\mathbb{P}}_i^{\text{logit}} + w_2 \hat{\mathbb{P}}_i^{\text{rf}} + w_3 \hat{\mathbb{P}}_i^{\text{xgb}}$$

with $w_1 + w_2 + w_3 = 1$, and i the dyads.

7.2 Results

The results of the rare event logit, non parametric (Random Forest and XGBoost) and *Ensemble* models help us to investigate the nature of conflicts to answer the third problematic: what characteristics make two countries more likely to initiate a physical conflict, and conditional on a conflict having already occurred, what factors determine whether it persists over time ? The main finding

of this section is double. First, conflict persistence is more predictable than conflict onset. Second, cyber-related variables are among the most important predictors of conflict persistence while they play a smaller role in onset prediction. We now document the evidence supporting these claims.

7.2.1 Onset versus Persistence

This section first documents the asymmetry in predictive performance between onset and persistence models through three complementary evaluation metrics, before focusing then on the role of specific predictors and out-of-sample forecasts.

Probabilities estimates For clarity and to avoid redundancy we do not report the estimates for each models but rather evaluate the models as a whole by reporting ROC curves, P-R curves and Brier scores.

The Receiver Operating Characteristic (ROC) curve enables to visualize the classification performance of a binary model. The Area Under Curve (AUC) summarizes this information in a single value. Each ROC curves and AUC are reported in [Figure 14](#).

The Precision-Recall curve (P-R curve) enables to visualize the trade-off between recall and precision. Recall measures the model’s ability to find all the positive events, and precision measures the accuracy of positive predictions. P-R curves and AUC are reported in [Figure 15](#).

For each model, we also compute the Brier score. It is a metric used to evaluate the accuracy of predicted probabilities. The lower the score, the better the model performance. The Brier score baseline corresponds to a forecast where the predicted probability is constant and equal to the unconditional event prevalence in the sample:

$$BS_{\text{baseline}} = p(1 - p)$$

where p denotes the unconditional event prevalence in the sample. The results of the Brier Score are reported in [Table 12](#).

ROC curves For the onset models we can observe in [Figure 14](#) that *Ensemble* performs better than the others, with an AUC of 0.653, followed by Random Forest (RF) with an AUC of 0.651, the rare events logit and XGBoost (XGB) with, respectively, an AUC of 0.602 and 0.601. However, as

a random model will have an AUC of 0.5, this indicates that even if the prediction power of these models are better than for a random assessment, they remain limited. This was expected because predicting a new conflict is complex as it depends on many factors that are difficult to capture.

From the ROC curves of the persistence models, very different results are observed. Indeed, the model with the best performance is the logit with an AUC of 0.754, then *Ensemble*, RF and finally XGB with AUC of 0.752, 0.748 and 0.705. For each of the models, the performance is well better than for the onset conflict. Again, this was expected as conflicts are already observed. Then, with the help of the lagged variables, the persistence is easier to predict.

P-R curves [Figure 15](#) illustrates the Precision-Recall curves. The dotted line represents the baseline scenario of onset and persistence. On the testing set, 7.6% of the observations are onset: a random model will have a precision of 7.6%. All the onset models perform better than a random one but they still perform poorly to predict onset. The corresponding AUC are 0.119 for the logit, 0.149 for RF, 0.115 for XGB and 0.150 for *Ensemble*.

A random model for persistence will have a precision of 35.2%. Persistence models perform very well in comparison to the onset ones. Indeed, the best models are RF (AUC = 0.684) and *Ensemble* (AUC = 0.680). In the RF case, when recall $\approx$ 50% and precision $\approx$ 80%, the model detects 50% of the persistence with 80% of precision.

Brier Scores [Table 12](#) illustrates the results for each models. For the onset models, only RF and *Ensemble* have a lower score than the baseline scenario, therefore these models predict probabilities slightly more precise than the baseline. However this improvement is marginal.

Similarly, for persistence models, both RF and *Ensemble* have the lower Brier scores. This indicates that both estimates are closer to the observations than the other models. Moreover, we can notice a clear difference in comparison to the baseline Brier score (0.1810 and 0.1820 vs 0.2281).

These three metrics confirm the asymmetry announced before: persistence is more predictable than onset across all models and all evaluation criteria, with RF and *Ensemble* achieving the strongest gains for persistence.

7.2.2 Models Results

Variables importance To understand what drives predictions, we look at the feature importance of the RF model. We focus on RF rather than XGBoost or logit because it achieves the best average performance across all metrics on both onset and persistence. [Figure 16](#) reports the twelve most important variables for each model.

For onset models, main features are demographic, political and economic. Indeed, the population size of the two countries is a very important factor of conflict. Then, the average Internet access rate in the dyad, the ratio of military power or peace duration are also important.

For the persistence model, the variables are very different. The most important one is war duration, the longer a conflict lasts, the more likely it is to continue. We can notice the importance of cyber-related features. Indeed, among the twelve most important variables, there are seven cyber-related ones. This points to the central role of cyber dynamics in sustaining ongoing conflicts.

Conflict forecasts Given that the *Ensemble* model is the one with the best performances, we use it to do forecast on the 2020 to 2024 period. The 2025 forward forecasts are performed by using a simple unweighted average of the three models, as it allows to reduce overfitting. Indeed, adapting the weight depends on the data, which could lead the model to adjust too much to it, and then perform poorly when forecasting. The forecast is therefore generated as follows:

$$\hat{\mathbb{P}}_i^{\text{forecast}} = \frac{\hat{\mathbb{P}}_i^{\text{logit}} + \hat{\mathbb{P}}_i^{\text{rf}} + \hat{\mathbb{P}}_i^{\text{xgb}}}{3}$$

with i the dyads.

To take into account the model uncertainty, we also report an uncertainty band. Since, the predicted probability is the mean of the three models, the lower band is the lowest predicted probability, and the upper band is the highest. Note that this is different from a confidence interval as it does not include any estimated distribution, estimated variance or confidence level.

Both backtesting results (2020-2024) and out-of-sample forecasts (2025) for our six target dyads are represented in [Figure 17](#). Red lines represent the predicted probability of conflict persistence, while blue lines represent the predicted probability of conflict onset. The dashed (red and blue) lines

correspond to the baseline mean probabilities across all dyads in the dataset: 35.2% for persistence and 7.6% for onset.

Backtesting (2020-2024). For the five persistence dyads, predicted probabilities are above the 35.2% baseline for the entire period. Russia-Ukraine, Iran-Israel, Iran-USA, and India-Pakistan, achieve a perfect accuracy of 100% for the testing period. Russia-USA reaches 75% accuracy, with a single false positive in 2021, where the model predicted persistence while the dyad was temporarily at peace (this can be seen by the cross on the graph). The weakest results is Afghanistan-Pakistan, where the model achieves 67% accuracy but fails to detect any true positive, likely reflecting the intermittent and hard-to-predict nature of this conflict. Full accuracy, precision and recall results for all dyads are reported in [Table 13](#).

Concerning new conflict, the Afghanistan-Pakistan dyad shows a predicted probability of 34% in 2021 which is well above the baseline of 7.6%. Hence the model correctly identified the onset that occurred that year. Then, the onset probability declined to 22% in 2024 where there was no conflict onset. For Russia-USA, the blue point at 44% (above the 7.6% baseline) in 2022 reflects the model correctly detected an elevated risk of direct confrontation, consistent with the sharp escalation in US-Russia tensions triggered by the Russian invasion of Ukraine that year.

Out-of-sample forecasts (2025). For most of the dyads, the model predicts high conflict probabilities persistence above the 35.2% baseline. Russia-USA is forecasted at 83% in 2025. Iran-USA at 82%, Russia-Ukraine at 77%, and India-Pakistan at 74%. Iran-Israel is forecasted at 51%, reflecting an increased uncertainty. This probability stay above the baseline but is lower than the 85-90% range observed during backtesting.

For Afghanistan-Pakistan, the model predicts an onset probability of 9% in 2025 which is low but remains above the 7.6% baseline.

Because this study is conducted in early 2026, these forecasts can be partially validated ex-post. All of our predicted high-risk dyads experienced major armed confrontations in 2025. The Iran-Israel conflict escalated into a direct military confrontation in June 2025 (the drop in forecast probability to 51% may reflect post-conflict ceasefire dynamics). The India-Pakistan dyad had an armed

confrontation in May 2025. For Iran-USA, US military involvement in support of Israel in June 2025 is consistent with the 82% persistence forecast. The Afghanistan-Pakistan dyad experienced cross-border armed battle in October 2025, consistent with the model's onset probability of 9%. This finding illustrates the model's ability to detect elevated risk even for lower-probability events. Finally, Russia-Ukraine and Russia-USA both remained in active conflict in 2025, consistent with persistence forecasts of 77% and 83% respectively.

7.3 Take away

Cyber variables emerge as central to understand conflict persistence. The six ex-post verifications (in 2025) provide strong evidence of the model's predictive validity at the dyadic level, and confirm that persistence which is driven by the self-reinforcing dynamics documented in Section 5.1 is empirically more predictable than onset. This partially supports Hypothesis 3 and suggests that cyber activity is more informative about the continuation of existing conflicts rather than the triggering of new ones.

8 Conclusion and limitations

This paper investigates the relationship between cyberattacks and armed conflicts through three complementary empirical approaches, combining dynamic analysis, economic impact assessment, and dyadic-level predictions.

First, the analysis of bidirectional dynamics shows the existence of interactions between cyber and physical conflicts. At the country level, results highlight strong persistence in both phenomena and show that physical conflicts significantly drive cyberattacks in the short run.

At the global level, the evidence points to a richer dynamic structure: while conflicts tend to precede cyber incidents, VAR estimates reveal a bidirectional Granger causality, indicating mutual short-term feedback effects.

In the long run, VECM results suggest that cyberattacks amplify deviations from equilibrium, whereas physical conflicts act as the adjustment mechanism restoring stability.

Second, the analysis of economic consequences provides more surprising results. Once physical conflicts are accounted for, cyberattacks appear positively associated with GDP per capita.

However, this counterintuitive finding is likely driven by reverse causality. At the same time, no robust direct effect of physical conflicts on GDP per capita is identified, which may reflect the difficulty of capturing localized economic disruptions using aggregate national indicators. These results highlight important identification challenges and suggest that future research should rely on causal inference strategies with instrumental variables to better isolate the economic impact of both types of conflict.

Finally, the dyadic analysis confirms that predicting the onset of conflicts remains extremely challenging, with all models performing marginally better than random benchmarks. In contrast, the persistence of conflicts is significantly more predictable, reflecting the strong temporal dependence of armed violence.

However, this paper contains several limitations. The first limit is the lack of perspective according to existing working papers. Indeed, the literature at the intersection of the three topics studied is very limited as the systematic study of cyber threats as a driver of geopolitical risk is still an emerging field. A second limit concerns the nature of the data, working with rare events make estimation and prediction very challenging. Third, cyber incident data may suffer from under reporting bias, or inconsistencies across countries and time. Fourth, concerning cyber events, we do not take into account the initiator and the target of the attack, we count the observation only as an interaction between countries whereas it could also have been a non-state affiliated attack. This reflects both a limitation and an expansion: a deeper exploration of the nature of relationships in the context of cyberattacks.

Nevertheless, studying economics, conflicts and cyber threats together remains very fascinating, and the topic offers many opportunities for further research. First, by extending this framework by using new and updated databases for cyber incidents to have a more complete overview of cyber threats around the world. In addition, new variables in the models could be introduced to account for omitted variables or used as instrumental variables as explained earlier. The major improvement would concern the study of the relations between countries to understand much better the nature of armed conflicts, the nature of cyber threats and finally the interaction between both.

To conclude, this paper provides consistent evidence that cyberattacks are an integral compo-

ment of modern warfare dynamics. While physical conflicts continue to structure the system in the long run, cyber operations increasingly influence escalation patterns. Even if introducing this paper by suggesting a future World War III may have been somewhat overstated, the empirical results point to a concerning reality: the persistence probabilities of current high-risk conflicts remain elevated.

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Appendix

Descriptive statistics

Table 1: Descriptive statistics – Country panel variables

Variable	Mean	SD	Min	P25	Median	P75	Max
GDP	17187.73	22073.72	243.08	2542.11	6626.26	26713.08	167187.20
Internet use (%)	45.16	32.83	0.00	12.31	44.10	76.11	100.00
Log conflict	0.255	0.645	0.00	0.00	0.00	0.00	5.204
Log cyber incidents	0.525	1.036	0.00	0.00	0.00	0.693	7.211
Log GDP per capita	-7.449	2.578	-14.151	-9.348	-7.470	-5.626	1.495
Mobile subscriptions	40076299	132593600	0	2132860	8257090	28849075	1824420000
Number of conflicts	0.969	4.908	0	0	0	0	181
Number of cyber incidents	3.919	26.867	0	0	0	1	1353
Population	52487401	167083400	32645	4310035	11017318	39074218	1450936000

Note: Log variables are computed as $\log(x + 1)$. The country panel covers the period 2000–2024.

Table 2: Descriptive statistics – Dyadic panel variables

Variable	Mean	SD	Min	P25	Median	P75	Max
Both EU	0.011	0.104	0	0	0	0	1
Both WTO	0.533	0.499	0	0	1	1	1
CINC ratio	0.005	3.057	-9.813	-2.164	-0.101	1.991	8.940
CINC total (log)	-3.191	1.227	-6.876	-4.092	-3.305	-1.889	-1.010
Common language	0.128	0.334	0	0	0	0	1
Conflict (binary)	0.059	0.236	0	0	0	0	1
Conflict onset	0.042	0.200	0	0	0	0	1
Conflict persistence	0.314	0.465	0	0	0	1	1
Contiguity	0.136	0.343	0	0	0	0	1
Cyber acceleration	0.049	3.745	-105	0	0	0	161
Cyber incidents (binary)	0.099	0.298	0	0	0	0	1
Cyber incidents (sum 3 years)	1.721	6.540	0	0	0	0	265
Cyber trend	0.217	3.645	-42	0	0	0	173
GDP per capita ratio (log)	0.129	1.899	-5.655	-1.110	0.165	1.530	5.081
Total GDP per capita (log)	10.064	0.972	6.825	9.288	10.284	10.863	12.088
Average internet use	45.034	26.920	0.024	22.240	44.450	67.151	100.000
Log cyber incidents (lag 1)	0.158	0.542	0	0	0	0	5.182
Log distance	8.182	1.021	4.304	7.511	8.275	8.996	9.854
Peace duration	8.911	6.649	0	3	8	14	24
Average polity	1.938	5.032	-10	-1	1.5	7	10
Polity difference	8.844	6.527	0	3	7	15	20
Trade asymmetry	0.440	0.831	0	0.058	0.169	0.465	13.209
Total trade (log)	13.555	3.162	0.113	11.881	13.979	15.692	20.743
War duration	0.112	0.724	0	0	0	0	14

Note: The dyadic panel covers the period 2000–2024. Binary variables indicate whether a characteristic applies to both countries in the dyad (e.g., both WTO members). Log variables are computed using $\log(x + 1)$ transformations.

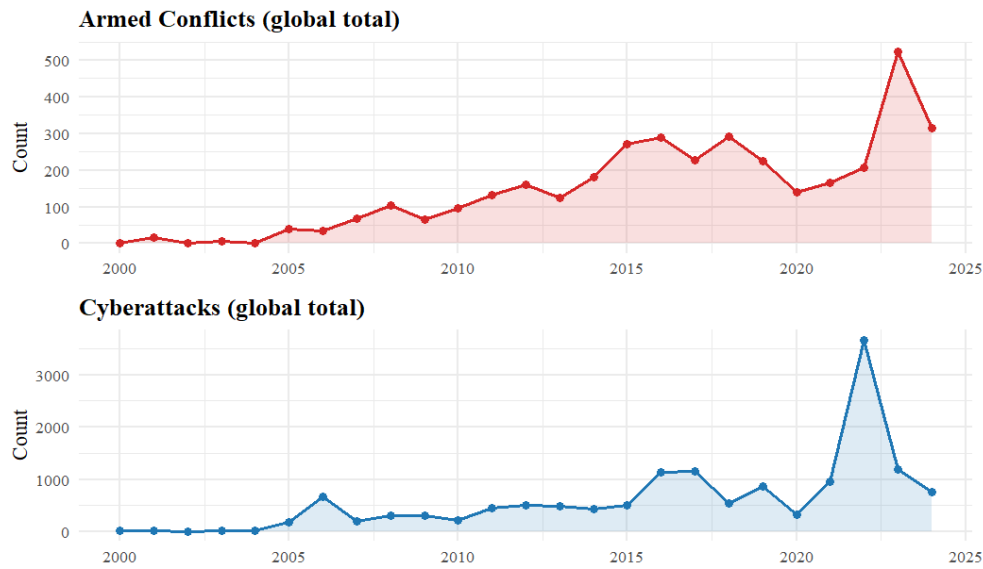


Figure 1: Global aggregate series (2020-2024)

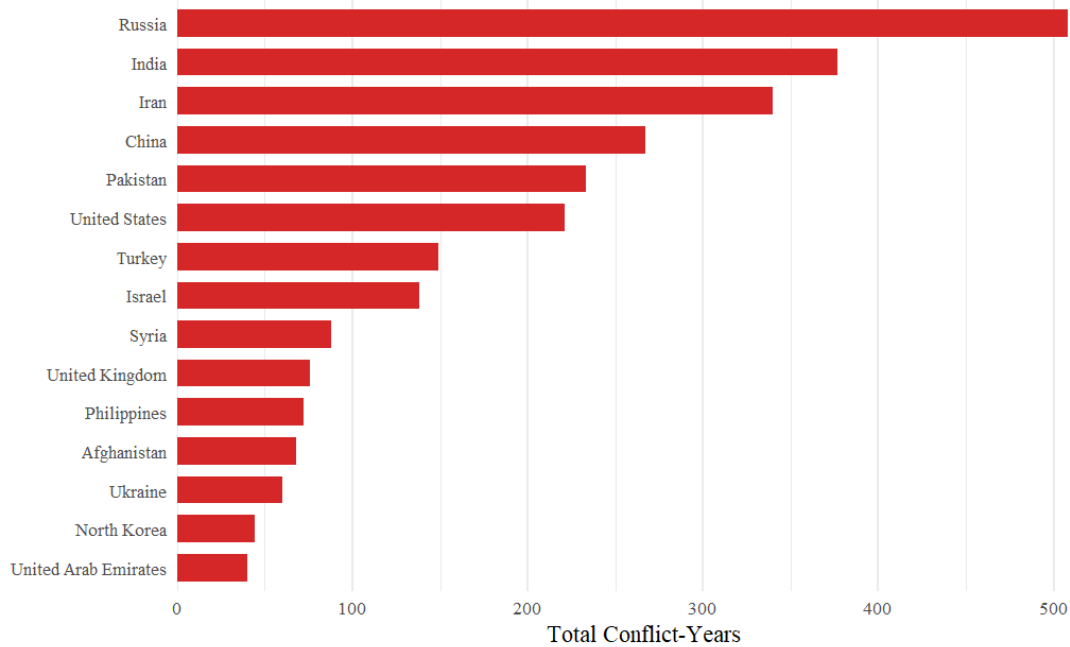


Figure 2: Top 15 conflict-prone countries (2020-2024)

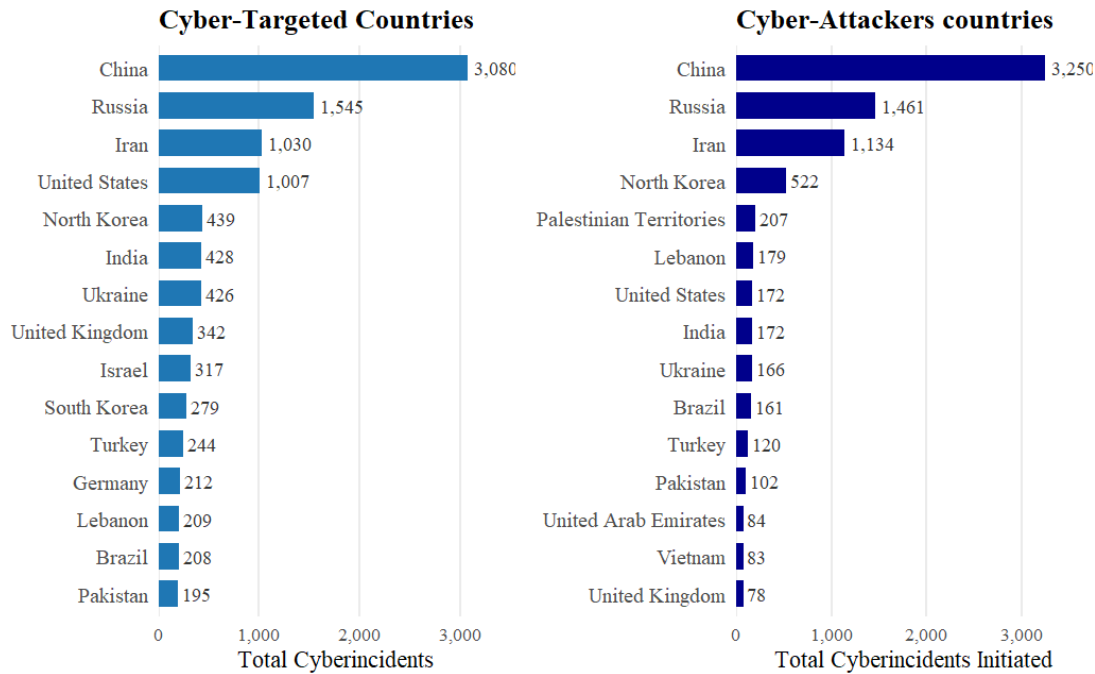


Figure 3: Top 15 Cyber-Targeted and Cyber-Attackers countries (2020-2024)

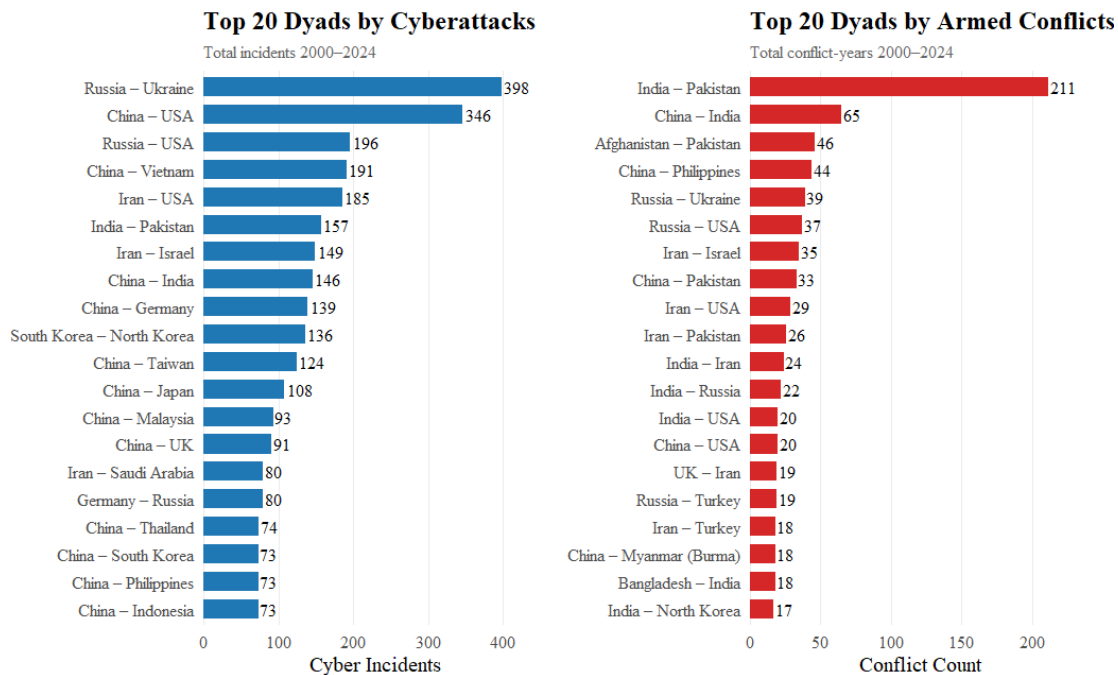


Figure 4: Top 20 dyads: Cyberattacks and Conflicts

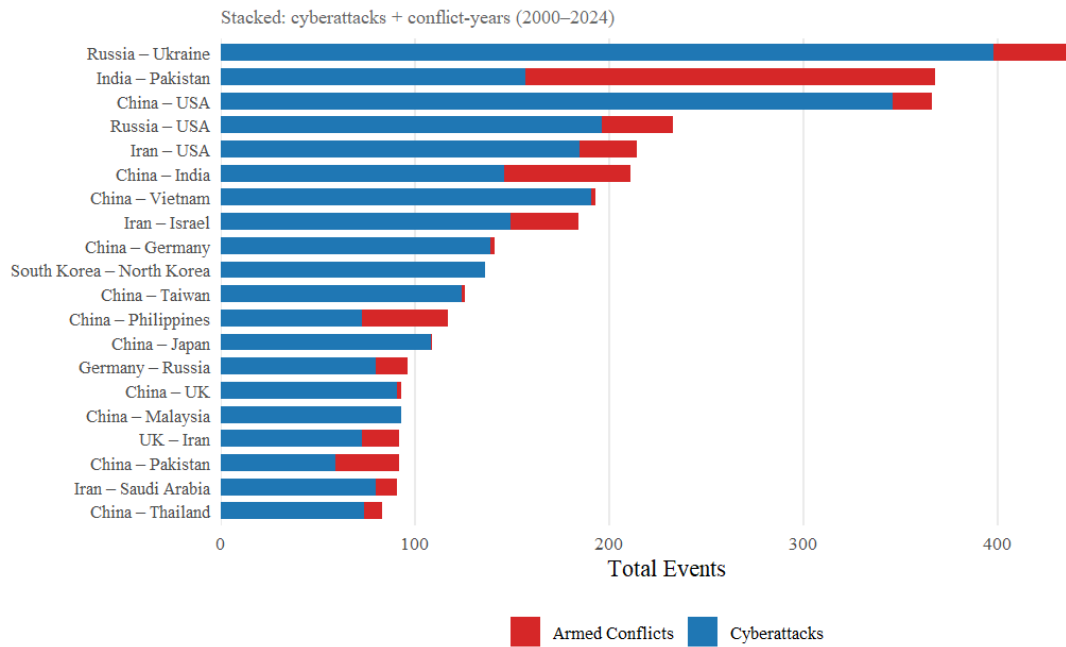


Figure 5: Top 20 dyads: Total events (Cyber + conflicts)

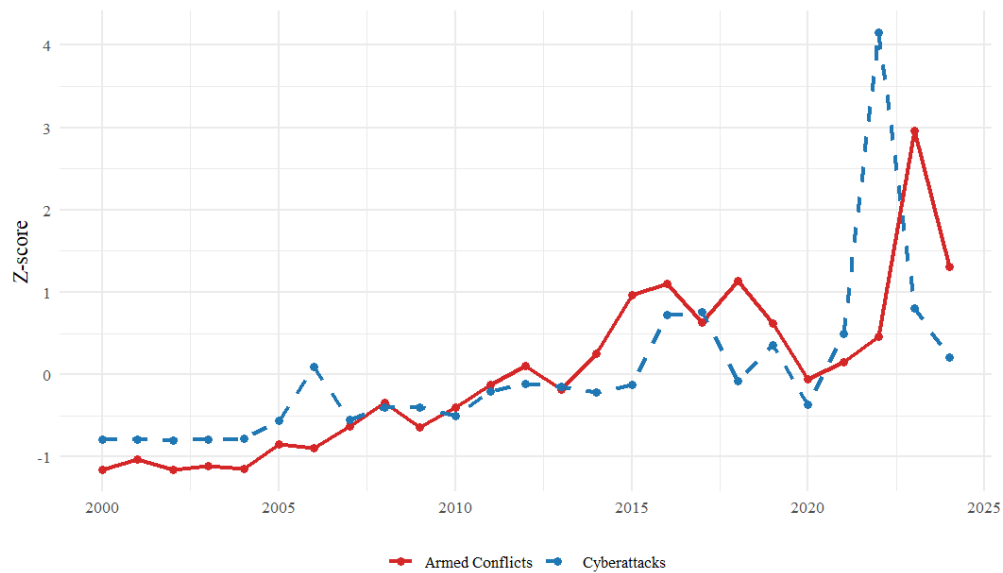


Figure 6: Standardized co-movement: conflicts and cyberattacks

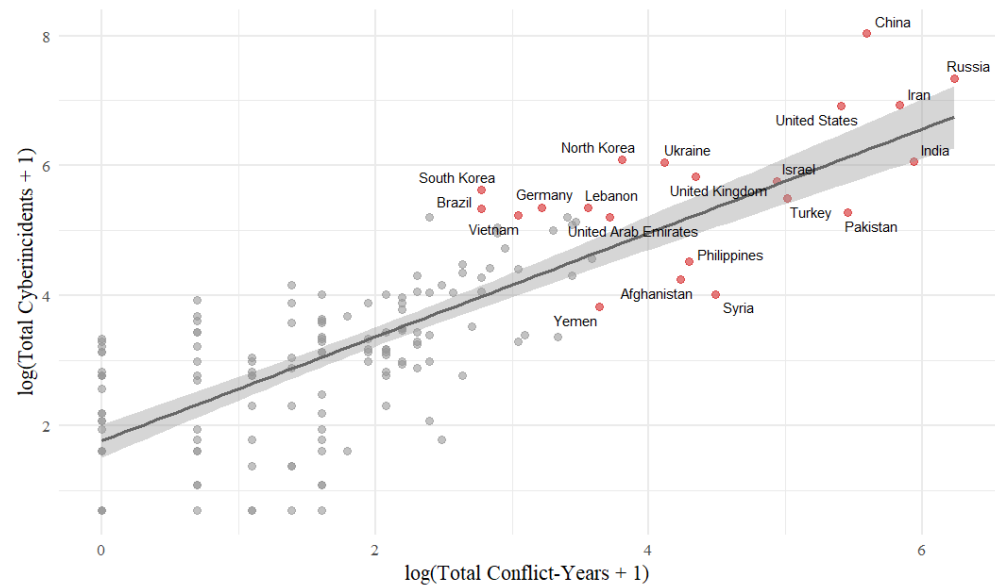


Figure 7: Cyber VS Conflict exposure (country-level)

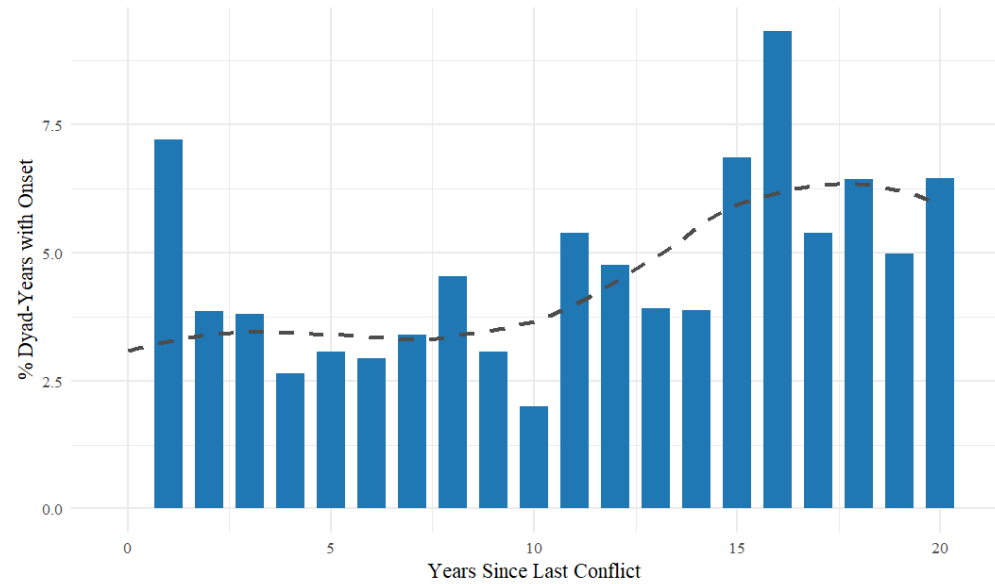


Figure 8: Conflict Onset Probability by Peace Duration

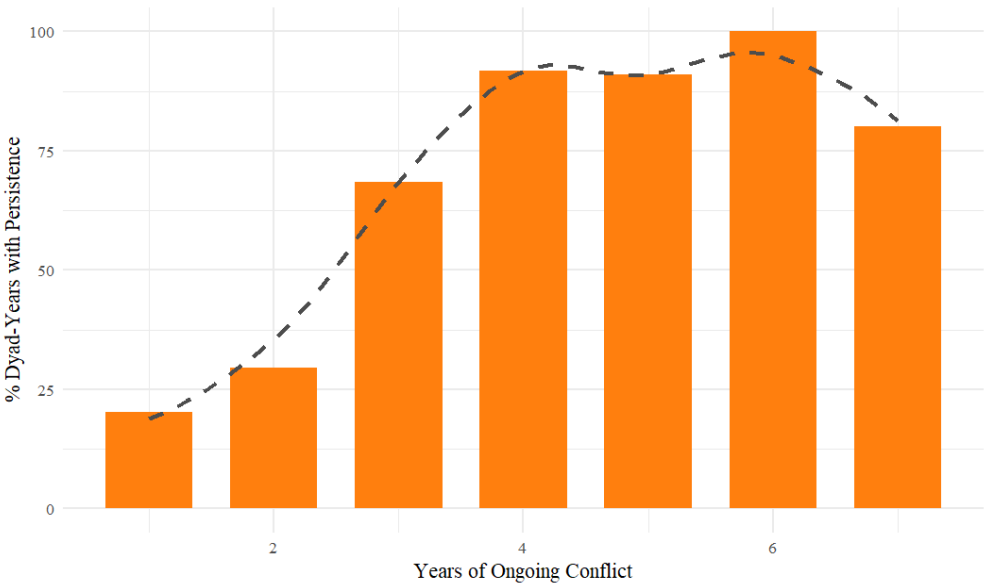


Figure 9: Conflict Persistence Probability by War Duration

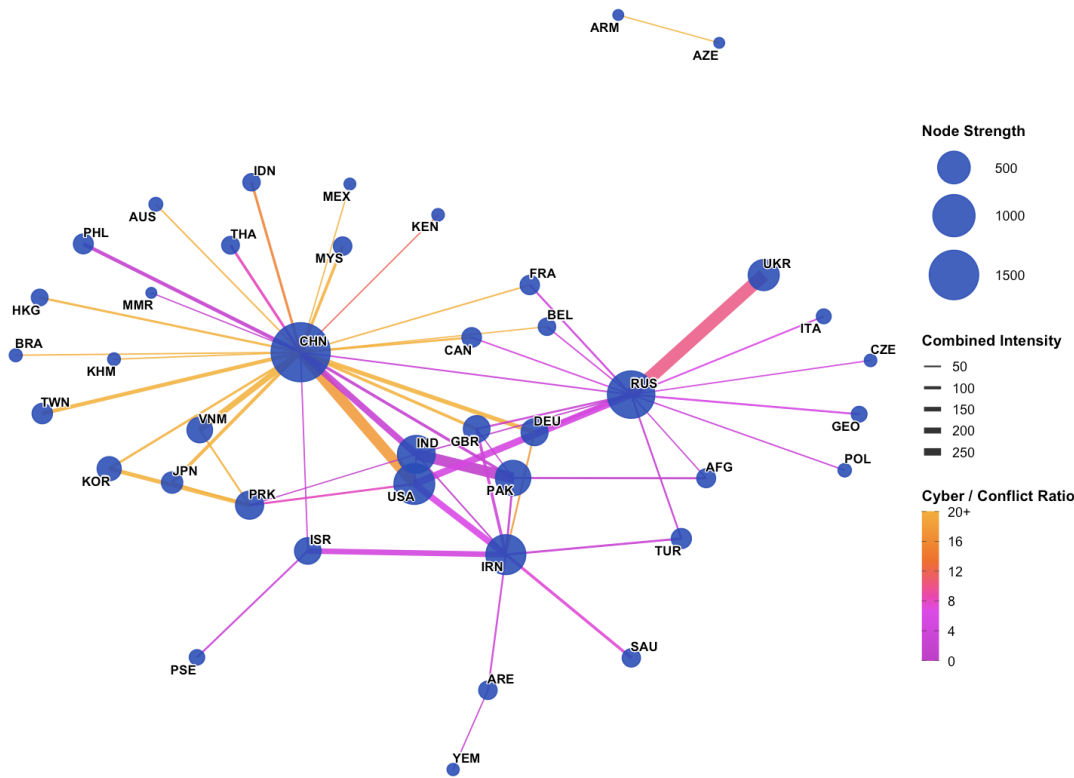


Figure 10: Combined network of conflicts and cyberattacks

Empirical strategy

Table 3: Covariates used in the conflict onset model

Category	Variable	Description
Conflict history	Conflict_lag2	Conflict in dyad i two years earlier
	Conflict_lag3	Conflict in dyad i three years earlier
Cyber activity	Peace_duration	Duration of the ongoing peace spell
	Cyber_lag1	Cyber incidents in $t - 1$
	Cyber_lag2	Cyber incidents in $t - 2$
	Log_cyber_lag1	Log of cyber incidents in $t - 1$
	Cyber_accel	Change between cyber activity at $t - 1$ and $t - 2$
	Cyber_sum3	Sum of cyber incidents over the previous three years
	Cyber_trend	Trend in cyber activity ($t - 1 - t - 3$)
	Cyber_x_tension	Interaction between cyber activity and prior tensions
	Cyber_x_polity	Interaction between cyber activity and regime differences
Political regime	Polity_diff	Absolute difference in Polity scores
	Polity_avg	Average Polity score in the dyad
	Polity_min	Minimum Polity score in the dyad
Power distribution	CINC_ratio	Log ratio of CINC capabilities
	CINC_total	Total capabilities in the dyad
Economic characteristics	GDP_ratio	Log ratio of GDP per capita between states
	GDP_total	Total GDP per capita of the dyad
	Internet_avg	Average internet penetration
	Log_pop1	Log population of country 1
Trade	Log_pop2	Log population of country 2
	Trade	Bilateral trade level
	Trade_total	Total bilateral trade flows
	Trade_asymm	Trade asymmetry between states
Institutions	Both_WTO	Both states members of WTO
	Both_EU	Both states members of EU
	Both_GATT	Both states members of GATT
Geography	Log_distw	Log weighted distance between states
	Contig	Contiguous states
	Comlang_off	Common official language
	Comrelig	Common religion
	Comcol	Shared colonial history
	Col45	Colonial link after 1945
	FTA_WTO	Free trade agreement
Controls	Year	Year fixed effects

Table 4: Covariates used in the conflict persistence model

Category	Variable	Description
Conflict dynamics	War_duration	Duration of the ongoing conflict
	Conflict_lag2	Conflict two years earlier
Cyber activity	Conflict_lag3	Conflict three years earlier
	Cyber_lag1	Cyber incidents in $t - 1$
	Cyber_lag3	Cyber incidents in $t - 3$
	Log_cyber_lag1	Log cyber incidents in $t - 1$
	Log_cyber_lag2	Log cyber incidents in $t - 2$
	Log_cyber_lag3	Log cyber incidents in $t - 3$
	Cyber_accel	Change in cyber activity between years
	Cyber_sum3	Sum of cyber incidents over previous three years
	Cyber_trend	Trend in cyber activity
	Cyber_x_tension	Interaction between cyber activity and tensions
	Cyber_x_polity	Interaction between cyber activity and regime differences
Political regime	Polity_diff	Absolute difference in Polity scores
	Polity_avg	Average Polity score in dyad
	Polity_min	Minimum Polity score in dyad
Power distribution	CINC_ratio	Log ratio of CINC capabilities
	CINC_total	Total capabilities in dyad
Economic characteristics	GDP_ratio	GDP ratio
	GDP_total	Total GDP of dyad
	Internet_avg	Average internet penetration
	Log_pop1	Log population of state 1
	Log_pop2	Log population of state 2
Trade	Trade	Bilateral trade level
	Trade_total	Total bilateral trade flows
	Trade_asymm	Trade asymmetry
Institutions	Both_WTO	Joint WTO membership
	Both_EU	Joint EU membership
	Both_GATT	Joint GATT membership
Geography	Log_distw	Log weighted distance
	Contig	Contiguity
	Comlang_off	Common official language
	Comrelig	Common religion
	Comcol	Shared colonial history
	Col45	Colonial link after 1945
	FTA_WTO	Free trade agreement
Controls	Year	Year fixed effects

Table 5: Hyperparameters of the machine learning models

Parameter	Random Forest	XGBoost
Number of trees	1000	—
<i>mtry</i>	$\lfloor \sqrt{p} \rfloor$	—
Minimum node size	10	—
Importance measure	Permutation	—
Prediction type	Probabilities	—
Random seed	42	—
Boosting rounds (<i>nrounds</i>)	—	300
Maximum depth (<i>max_depth</i>)	—	4
Learning rate (η)	—	0.05
Subsample	—	0.8
Column subsample (<i>colsample_bytree</i>)	—	0.8
Minimum child weight	—	1
Gamma	—	0
Class imbalance correction	—	$scale_pos_weight = \frac{n_{neg}}{n_{pos}}$
Objective	—	binary:logistic
Evaluation metric	—	AUC

Empirical results

	$Conflict_{i,t}$	$Cyber_{i,t}$
$Conflict_{i,t-1}$	0.497*** (0.050)	0.838*** (0.189)
$Cyber_{i,t-1}$	0.037 (0.038)	0.283*** (0.085)
Constant	0.284*** (0.079)	1.307*** (0.252)

Table 6: Panel VAR Estimation Results (System GMM, Two-Step)

Equation	Variable	Estimate	Std. Error	t value	$Pr(> t)$
Conflict	conflict.l1	0.403	0.249	1.616	0.125
	cyber.l1	0.088	0.019	4.607	< 0.001***
	conflict.l2	-0.175	0.251	-0.700	0.493
	cyber.l2	-0.025	0.029	-0.851	0.406
	const	-14.234	30.384	-0.468	0.645
	trend	6.884	4.243	1.622	0.123
$R^2 = 0.877$, $\bar{R}^2 = 0.841$, $F(5, 17) = 24.18$, $p < 0.001$					
Cyberattacks	conflict.l1	2.610	2.373	1.100	0.287
	cyber.l1	-0.188	0.181	-1.035	0.315
	conflict.l2	-7.907	2.383	-3.317	0.004**
	cyber.l2	-0.746	0.277	-2.691	0.015*
	const	-921.615	289.006	-3.189	0.005**
	trend	192.754	40.361	4.776	< 0.001***
$R^2 = 0.689$, $\bar{R}^2 = 0.598$, $F(5, 17) = 7.543$, $p < 0.001$					

Table 7: VAR(2) estimation results for global conflicts and cyberattacks (sample size: 23, log-likelihood: -289.235). Significance codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	Estimate	Std. Error	t value	$Pr(> t)$
$\log_Cyber_{it}$	0.008	0.009	0.855	0.394

Table 8: Two Way Fixed Effect Panel Models without controls

	Estimate	Std. Error	t value	$Pr(> t)$
$\log_Cyber_{it}$	0.005	0.008	0.571	0.569
$Internet_{i,t-3}$	0.002	0.001	1.910	0.058*
$\log(mobile)_{i,t-3}$	0.0705	0.013	5.313	< 0.01***
$\log(pop)_{i,t-3}$	-1.589	0.114	-13.937	< 0.01***

Table 9: Two Way Fixed Effect Panel Models with controls

	Estimate	Std. Error	t value	$Pr(> t)$
$\log_Cyber_{it}$	0.018	0.008	2.157	0.033*
$\log_Conflit_{it}$	-0.024	0.019	-1.261	0.209

Table 10: Two Way Fixed Effect Panel Models Extended without controls

	Estimate	Std. Error	t value	Pr(> t)
$\log_Cyber_{it}$	0.013	0.006	2.012	0.046**
$\log_Conflit_t$	-0.019	0.017	-1.148	0.253
$Internet_{i,t-3}$	0.002	0.001	1.881	0.062*
$\log(mobile)_{i,t-3}$	0.071	0.013	5.346	< 0.01***
$\log(pop)_{i,t-3}$	-1.589	0.113	-14.032	< 0.01***

Table 11: Two Way Fixed Effect Panel Models Extended with controls

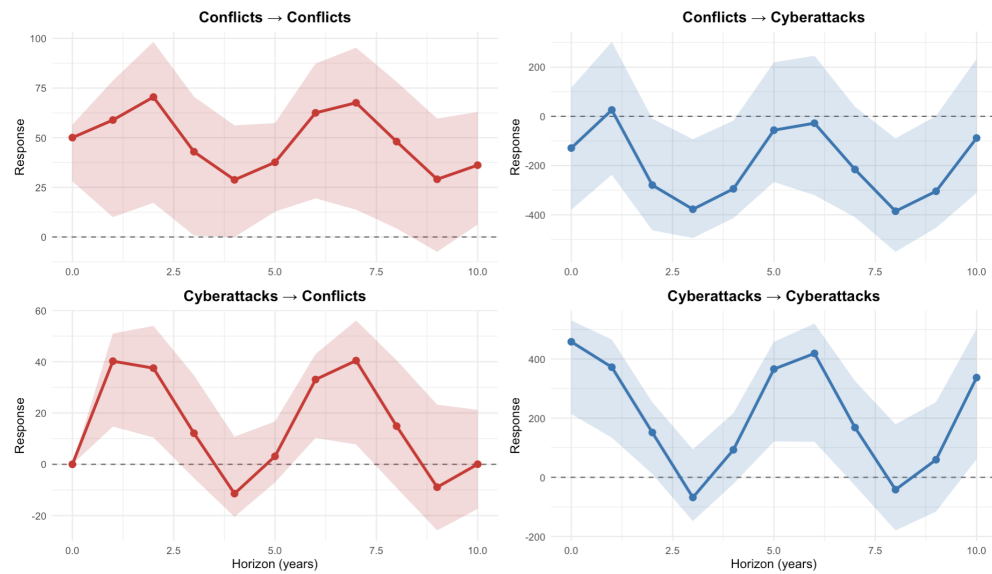


Figure 11: Impulse response function (IRF)

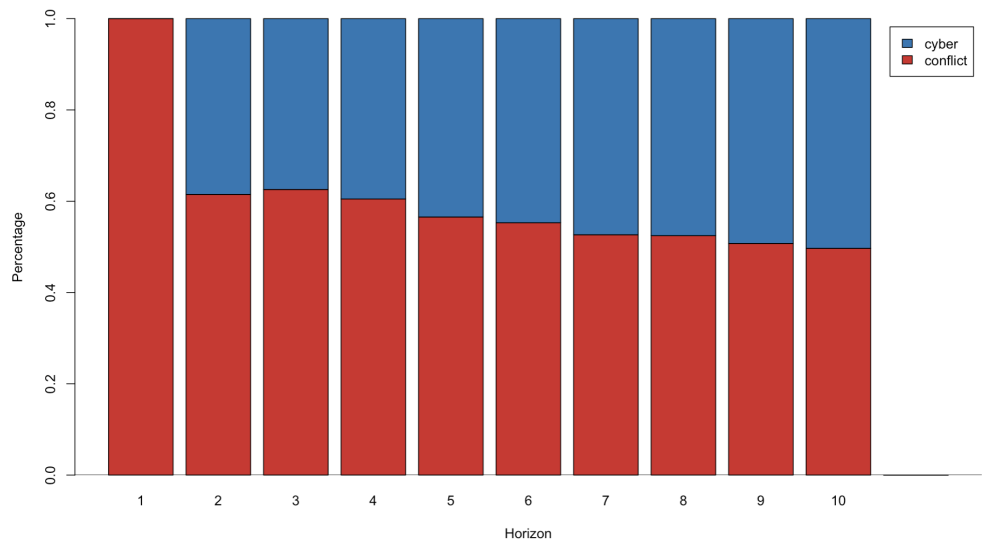


Figure 12: Forecast Error Variance Decomposition Conflict

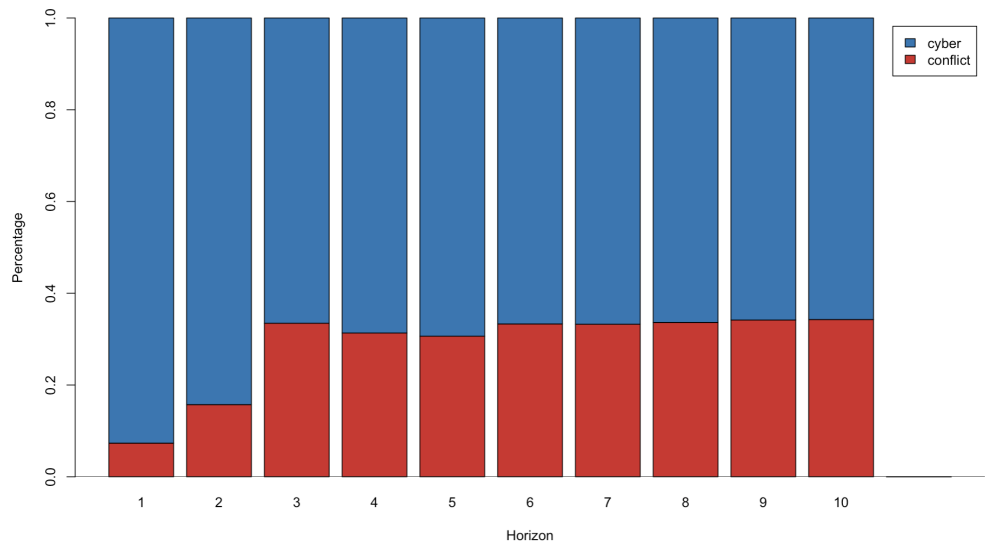


Figure 13: Forecast Error Variance Decomposition Cyber

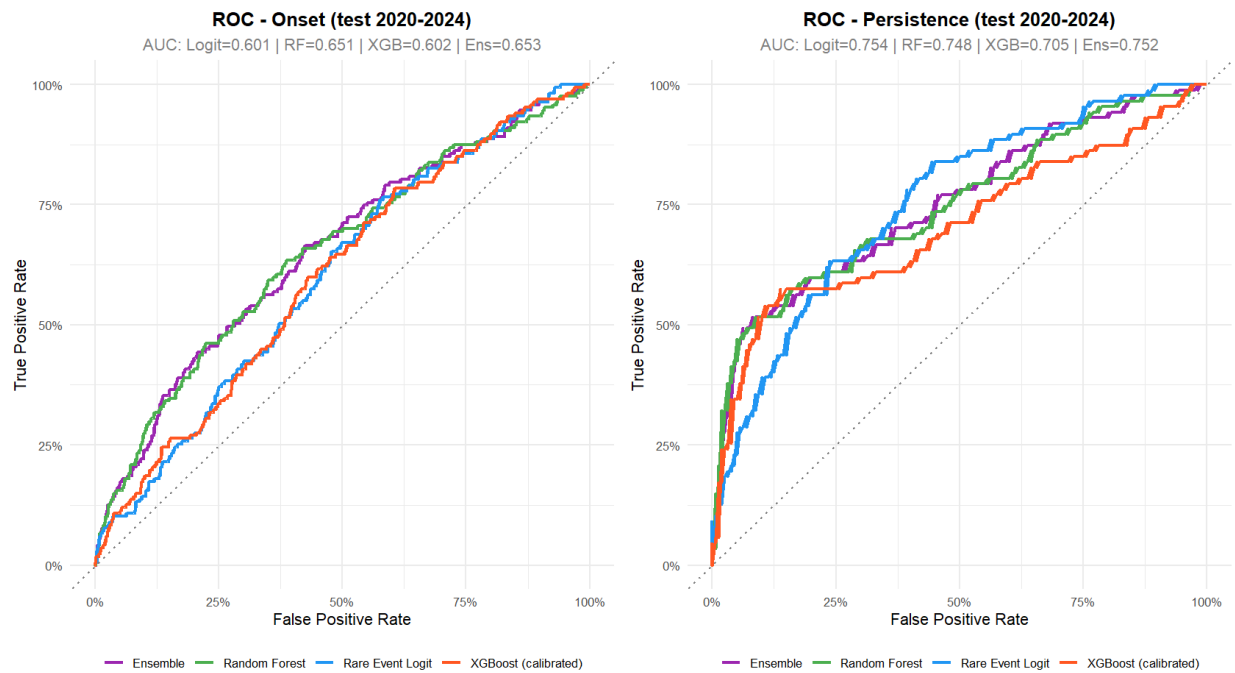


Figure 14: Roc Curves: Onset and Persistence models

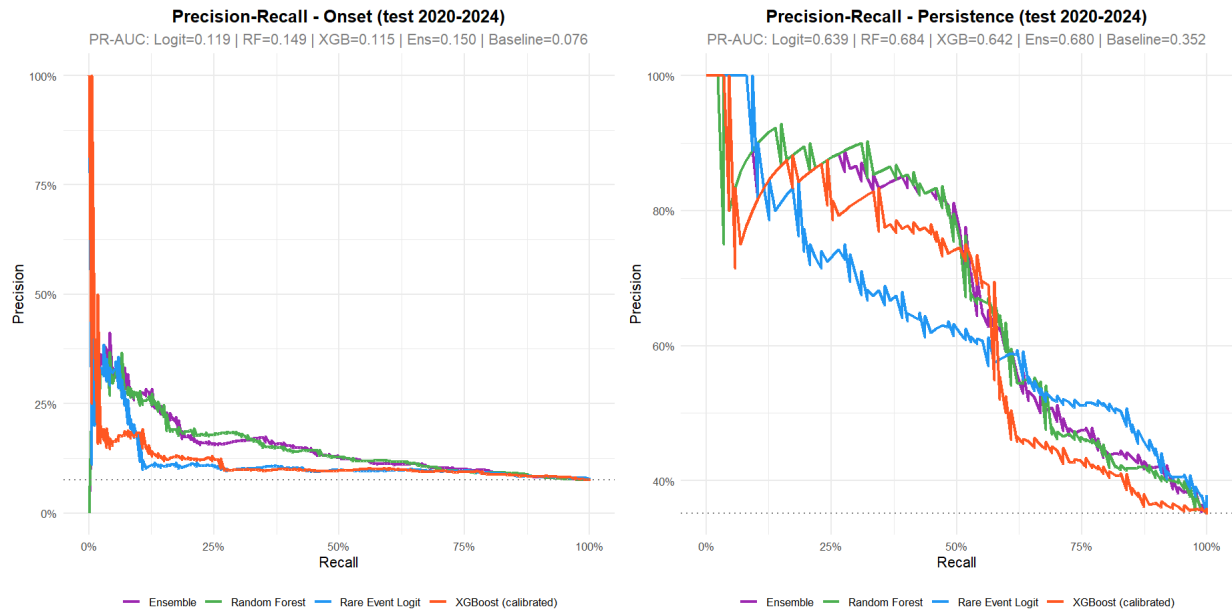


Figure 15: Precision-Recall Curves: Onset and Persistence models

Model	Onset	Persistence
Baseline (event prevalence)	0.0702	0.2281
Rare Event Logit	0.0808	0.2139
Random Forest	0.0701	0.1810
XGBoost (calibrated)	0.0731	0.2248
Ensemble	0.0695	0.1820

Table 12: Brier scores for conflict onset and persistence models (test set 2020-2024)

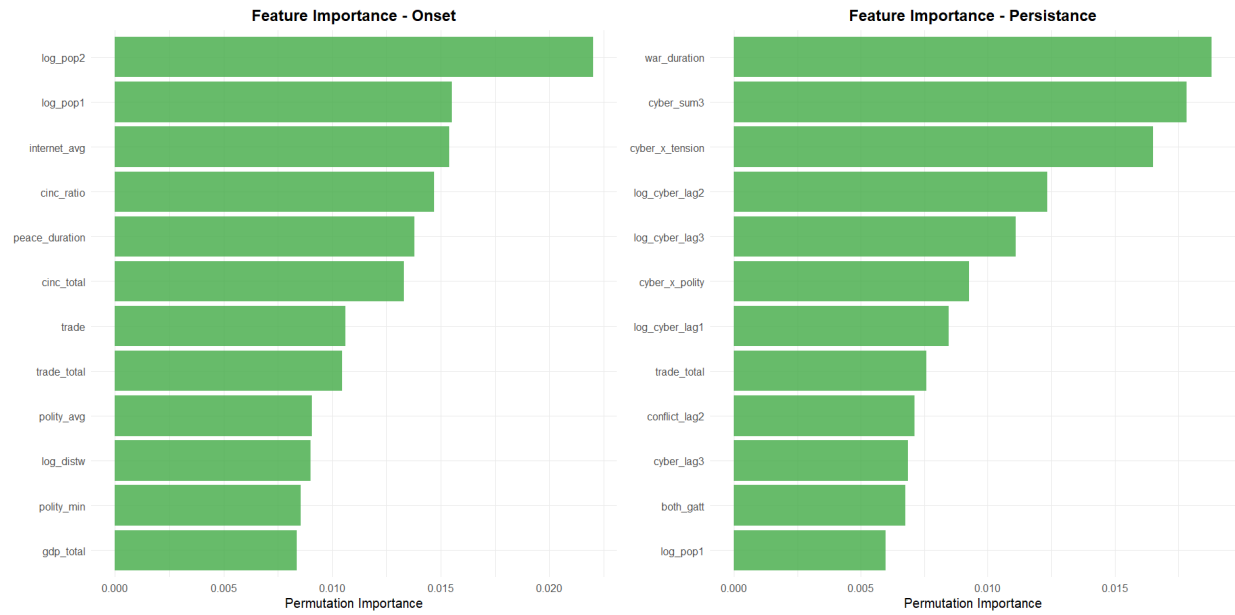


Figure 16: Permutation-based feature importance from the Random Forest models

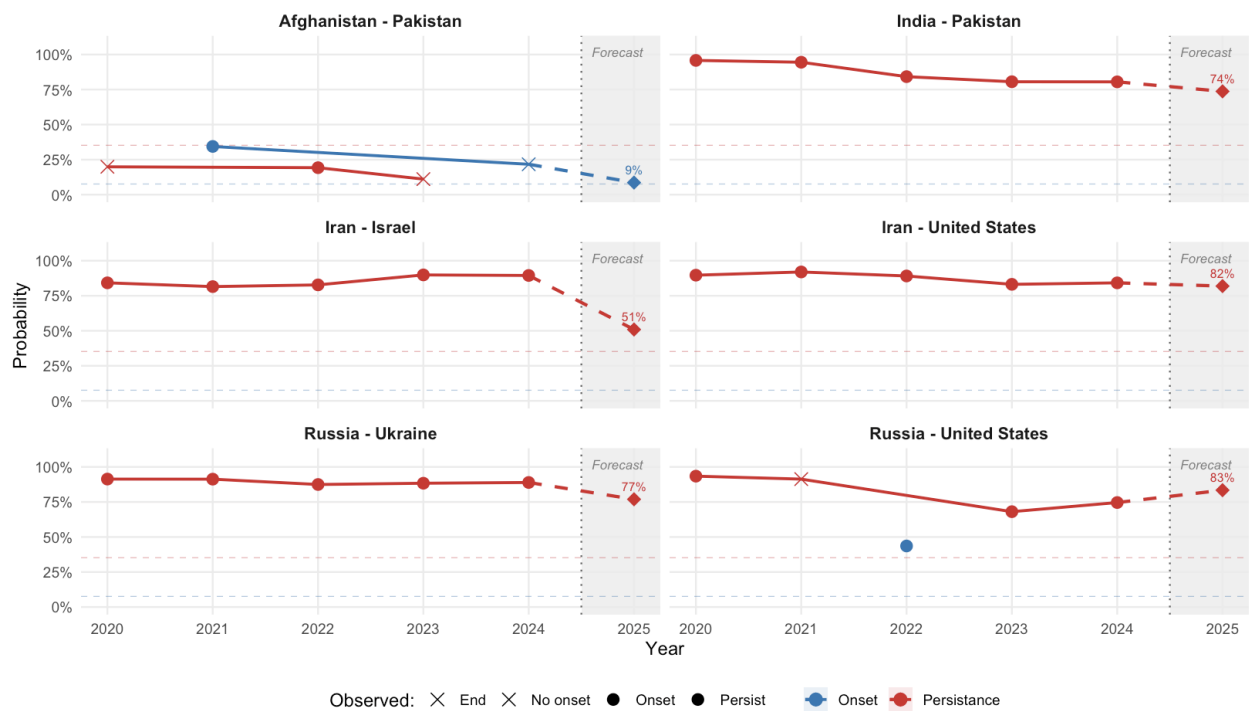


Figure 17: Back-testing (2020-2024) and Forecast (2025)

Table 13: Backtesting Accuracy by Dyad (2020–2024)

Dyad	Type	N obs	Accuracy	Precision	Recall
<i>Persistence</i>					
Russia – Ukraine	Persistence	5	100%	100%	100%
Iran – Israel	Persistence	5	100%	100%	100%
Iran – United States	Persistence	5	100%	100%	100%
India – Pakistan	Persistence	5	100%	100%	100%
Russia – United States	Persistence	4	75%	75%	100%
Afghanistan – Pakistan	Persistence	3	67%	0%	0%
<i>Onset</i>					
Russia – United States	Onset	1	100%	100%	100%
Afghanistan – Pakistan	Onset	2	50%	50%	100%

Note: Accuracy is computed relative to a threshold of $1.5\times$ the baseline prevalence. Russia–United States onset results are based on a single observation and should be interpreted with caution. The low precision and recall for Afghanistan–Pakistan persistence reflect the atypical and intermittent nature of this conflict.